Does Goliath Help David?
Anchor Firms and Startup Clusters

by

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Abstract

This paper investigates the effects of a large firm's geographical expansion (anchor firm) on local worker transitions into young firms through wage effects in industries economically proximate to the anchor firm. Using hand-collected data matched to administrative Census microdata, I exploit anchor firms' site selection processes to employ a difference-in-differences approach to compare workers in winning counties to those in counterfactual counties. The arrival of an anchor firm induces worker reallocation towards young firms in industries linked through input-output channels by a magnitude of 120 new businesses that account for approximately 2,300 jobs. Consistent with the literature in personnel and organizational economics, incumbent firms experiencing the fastest wage growth due to these shocks shed mid-layer employees who select into young firms within the county and in their own industry of experience. These effects are strongest in the most specialized and knowledge-intensive industries. Attracting an anchor firm to a county appears to have limited spillover effects in overall employment that are mainly driven by reorganization of incumbent firms in the anchor's input-output industries that face rising labor costs.

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1 Introduction

Given the importance of agglomerations on firm productivity and worker outcomes, one important question policymakers confront is whether attracting large firms induces employment and wage gains for the county as a whole. In particular, this paper asks whether large firms serve as an anchor for industry agglomerations by enabling the reallocation of labor across firms at the extensive margin. Programs geared towards attracting large firms to a municipality constitute a key component of many place-based policies aimed at inducing industry cluster formation, though these policies have often had mixed results (Chatterji, Glaeser, and Kerr 2014). In the hopes of becoming the next Silicon Valley, local policymakers often view attracting large firms as a means to jump start local economic activity and induce follow-on business formation. If this is the case, then geographic expansions of large firms can serve as "anchors" for new businesses in the county.[1] The arrival of an anchor firm may also induce increases in the cost of local factors, particularly in the short-run. These factor price changes may force some extant firms in the county-industry to reorganize and shed layers which may have implications far different from a story of entrepreneurs capturing entry opportunities unattainable by incumbent firms.

Workers may depart incumbent firms to start a new venture for a variety of reasons. One view is that large firms incubate employees who eventually spin-out to form new ventures by equipping them with technical expertise, nontechnical knowledge on regulatory and marketing strategies, and access to financiers (Gompers, Lerner, and Scharfstein, 2005; Chatterji, 2009). Another strand of literature uses the lack of en-

[1] I refer to the geographic expansions to a county by a large firm as an anchor firm. These geographic expansions are represented by the opening of a new facility by the anchor firm. Postdating this study, an example of an site search and anchor firm is DXC Technology's 2019 announcement for the opening of a 2,000 employee software publishing hub in New Orleans, LA. This facility is considered to be a possible "anchor firm" for software publishing in New Orleans.

trepreneurship in historically manufacturing-dependent cities as evidence that large capital-intensive firms deplete resources required by entrepreneurs (Chinitz, 1961; Jacobs, 1970; Glaeser, Kerr, and Kerr, 2015).[2] Reconciling the two views is empirically difficult for two reasons. First, data limitations have restricted analyses to a narrow set of industries. Second, studies discussing entrepreneurship and agglomeration face the key identification challenge posed by the endogeneity between a firm's location decision and the location's entrepreneurial climate.[3]

I demonstrate that the arrival of a large anchor firm induces startup formation specifically in its own supply chain. New ventures are more likely to be spinoffs from high quality incumbent supply chain firms.[4] In line with the role of a founder's industry and occupational knowledge in startup performance, these new startups grow faster and fail less than startups prior to the anchor firm's arrival. Comprehensive employee-employer matched Census microdata allows me to overcome typical data limitations and describe both the spinoff process and the shock that a large anchor firm's arrival provides to a local supply chain. This restricted-use data covers all businesses and workers in the United States regardless of industry since 1990. Firm and establishment data come from the Longitudinal Business Database (LBD) which represents the universe of over 200,000,000 establishment-year observations. The LBD allows me to observe outcomes for anchor firms, incumbent firms, and startups. To construct lifetime work histories on startup founders, I match the LBD to the Longitudinal Employer Household Dynamics (LEHD) data that includes 700,000,000 worker-job-year observations.[5]

[2] Agrawal et al. (2014) find that cities with a mix of small firms and large incumbent labs are most innovative by facilitating spinoffs.

[3] The identification challenge also inhibits isolating precise mechanisms describing how collocation with large firms and industry agglomeration affect new venture formation and performance.

[4] I refer to "incumbent" firms as firms that already exist in the county prior to the anchor firm's arrival.

[5] Section 3 describes identification of founders in the LEHD using firm age and wage rank within

To address endogeneity, I leverage instances where a large firm conducted a site selection process and revealed its leading alternative sites for a new facility. *Site Selection* magazine is corporate real estate publication that often publishes articles describing site searches of large, generally Fortune 500 firms.[6] The 240 site searches I document describe facilities usually employing over 500 employees and requiring an investment of a quarter billion dollars.[7] Table 1 displays the industry breakdown of the new facilities in the magazine and Table 1b shows basic characteristics of parent firms in the *Site Selection* sample. Observing that winning and runners-up counties are similar on dimensions related to entrepreneurial formation and industrial composition (Table 2), I obtain plausibly exogenous variation in startup formation and employ a difference-in-differences design to compare startups and their founders in winning with those in runners-up counties.[8]

Traditional Marshallian agglomerations form through a variety of channels including natural advantages, labor pooling, proximity to input-output markets, and knowledge spillovers through proximity. This paper focuses on the latter two channels with a particular emphasis on economic linkages between industries in both shipments across as well as labor similarity between them. I test the importance of such Marshallian channels using a difference-in-differences identification strategy that uses the introduction of a large anchor facility as a shock to local labor markets. The anchor firm's arrival serves as an unexpected shock to the demand for labor in firms geo-

a firm.

[6]These new openings by large firms in the data are "anchor firms."

[7]Greenstone, Hornbeck, and Moretti (2010) provide an overview of the magazine. The 240 site searches reflect *Site Selection* events for which I find company name, winning site, runners-up site, and a limited description on the facility's operations between 1990 and 2015. Not all of the 240 site searches were matched into the administrative Census data.

[8]For ease of exposition, I refer to these large firms conducting the site search as "anchor firms." "Winning counties" are the counties that the anchor firms ultimately decide to locate. Sites considered at the end of the site search process but not selected are "finalist" or "runners-up" counties. Greenstone et al. provide additional statistics on county similarity.

graphically proximate to a potential buyer or supplier of intermediate parts. Prior literature such as Ellison, Glaeser, and Kerr (2010) establish the coagglomeration patterns of input-output industries. Physical proximity can enable firms to better monitor their suppliers or tailor products to their customers. Highlighting the importance of proximity to the supply chain for startups, I find that the effects of the anchor firm's arrival on startup formation are localized to industries upstream and downstream to the anchor firm.[9] The average county-industry pair (defined at the NAICS-4 level) in the anchor firm's supply chain generates 22 more new startup business establishments representing 430 jobs.[10] For the average winning county, this amounts to over 120 new establishments from startup firms and 2,300 jobs.[11] Consistent with prior literature on the agglomeration of knowledge-intensive industries, the most pronounced effects on supply chain entry and startup job creation are in knowledge intensive supply chains. Examples of knowledge intensive supply chains include R&D Services, Semiconductors, and Aircraft Parts Manufacturing. Overall, supply chain startups in winning counties grow 12% faster in terms of employment and exit 7% less relative to those in runners-up counties after the anchor firm's arrival.[12]

This paper's most novel contribution is showing that the anchor firm's arrival simply reallocates labor along the firm age distribution potentially through the movements of mid-level managers who are pushed out from firms with rising wage bills. Workers above the median wage, but generally not the firm's top earners, depart surviving incumbent firms and form new ventures in economically proximate industries

[9]The supply chain is defined for each of the 240 site searches and consists of upstream supplier industries, the focal industry, and downstream buyer industries. These designations are defined using BEA Use of Commodities Tables.

[10]For each county, there can be up to 5 upstream and downstream industries.

[11]These totals amount to approximately 20-23% of the total number of supply chain establishments and employees across all age groups.

[12]Growth and failure are tabulated over the first five years of a startup's life. Startups in winning counties perform significantly better post-anchor firm than prior to the anchor firm's arrival as well.

that share supply chains and occupational similarity as their former employers.[13] After the anchor firm arrives, the number of employees departing an employer to start a new venture in the employer's input or output industries increases by almost 50%. As a share of all new supply chain startups, my coefficients imply that 25% of upstream startups are spinoffs as are 91% of anchor industry startups and 71% of downstream startups. These magnitudes are particularly striking and amount to 3 to 4 times the spinoff rate of these industries at the national level.

The formation of new ventures through employee departures from incumbent industry firms may not be driven exclusively by new creative entry opportunities. Instead, rising labor costs force firms to reorganize potentially through layer-shedding (Caliendo, Monte, and Rossi-Hansberg, 2015).Firms that reorganize shed layers and these newly pushed out workers form businesses in industries that most suit their skills. In line with this prediction, I find that firms with the fastest wage growth and highest average wage also generate the most spinoffs. A one standard deviation increase in an upstream firm's average pay relative to its industry increases the number of spinoffs by 10%. For anchor industry firms, a one standard deviation increase in average firm pay increases spinoffs by 21%. The effect on downstream firms is 17%.

1.1 Related Literature

I contribute to literature that describes agglomeration spillover effects and the geography of supply chains. Describing coagglomeration patterns, Ellison, Glaeser, and Kerr (2010) show that industries linked through intensity of shipments to each other as inputs and outputs tend to collocate. However, less clear is why this trend continues particularly as transportation costs continue to fall. One explanation could be

[13]In this paper, a spinoff is defined as employees who leave an incumbent firm and start a new company in the same industry category as their employer. Incumbent firms that generate spinoffs are referred to as "parent" firms.

that geographic proximity to buyers and suppliers enables better monitoring of transactions across segments of the supply chain and allows for a greater ability to tailor specialized services and products to downstream firms. These forces are greatest in industries related to R&D (Alcacer and Delgado, 2016) and I show that the mechanism is also relevant for knowledge intensive supply chains. Density of local labor markets is an additional channel discussed as a source of agglomeration benefits in clusters. The anchor firm's arrival reflects an increase in labor market density (Greenstone, Hornbeck, and Moretti, 2010), and in particular labor markets most related to the anchor firm's supply chain. Indeed, wages and within firm pay variance increase substantially in both the anchor industry as well as the most economically connected industries to the anchor industry. The literature in personnel economics (Lazear and Oyer, 2004; Lazear and Shaw, 2007) and organizational economics (Caliendo, Monte, and Rossi-Hansberg, 2015) discusses the balance between firm compensation policies and the number of occupations or layers within the firm. However, this literature is not typically connected to the literature in agglomeration. I connect to this latter literature by discussing that the gains to firm growth through collocation in related industries can be identified through the transition of workers who might be pushed out of firms rather than individuals who are necessarily high productivity types. This means that the fast growth rate and increased probability of survival is driven by agglomeration spillovers and not selection of high type individuals into entrepreneurship in dense labor markets (Combes et al., 2012).

This paper also contributes to literature on spinoffs that has shown incumbent firms to be incubators of future entrepreneurs (Gompers, Lerner, and Scharfstein, 2005; Chatterji, 2009). The extant literature points to two primary explanations for spinoffs. One reason employees spin-out is that they recognize market opportunities arise that their employer is unable to capture or the employer is unable to appropri-

ate returns to employees who discovered them (Klepper and Sleeper, 2005; Franco and Filson, 2006; Klepper, 2007; Klepper and Thompson, 2010). Another explanation is that some incumbent firms provide employees with networks and nontechnical expertise about the market and equip them for growth in knowledge-intensive product markets (Chatterji, 2009). However, unlike prior literature I demonstrate that shocks geographically proximate and in the employer's supply chain induce employee spinoffs. I further show that spinoffs may not necessarily be driven by star employees capturing returns bureaucratic incumbent firms cannot. Instead, spinoffs may also arise in agglomerations where workers are pushed out of firms that must reorganize to adapt to externally driven shocks to their wage structure. These employees are likely to select into firms or start firms in industries that they are most familiar which is plausibly that of their recent employer.

1.2 Organization

The remainder of this paper is as follows: Section 2 discusses possible mechanisms describing the process of spinoffs and agglomeration with a discussion of related literature. The extensive data used in this paper is detailed in Section 3. Section 4 explains the empirical methodology and Section 5 discusses results. I describe robustness of measurements as well as present evidence of subsidy similarity in 6. Finally, I conclude this chapter in Section 7 summarizing my main findings and describing avenues for future research.

2 Conceptual Framework on Spinoffs and Agglomeration

2.1 Spillovers versus Reorganization

This paper relates closely to two studies examining the spillover effects onto incumbent firms when an anchor firm arrives to a county. Greenstone, Hornbeck, and Moretti's (2010) are the first to use new establishment openings of large firms from Site Selection magazine and provide the basis for this paper's identification strategy. Focusing on manufacturing plants in the 1980's and early 1990's identified in the Census of Manufactures, they show that productivity of a county's incumbent plants in the same industry as the opening establishment increases. Their model assumes that the arrival of the anchor firm leads to an increase in the number of employers in an industry. They attribute productivity gains to enhanced labor productivity through knowledge spillovers arising from increased thickness in local markets.

Contemporaneous to this paper, Bloom et al. (2019) focus on manufacturing firms in the Annual Survey of Manufactures matched to their own *Site Selection* magazine sample spanning 2010-2014. Bloom et al. suggest that one component of Greenstone et al.'s knowledge spillovers mechanism is that the anchor firm induces better management practices in the county's incumbent manufacturing firms. These improvements may be reflected in productivity gains in incumbent firms.

While this paper uses its own *Site Selection* sample, it covers a different time horizon (1990-2015) and matches to employee-employer data. In addition, this paper expands beyond manufacturing plants and uses the movement of workers in the LEHD to discuss knowledge and worker ability. Prior works focus on outcomes at the intensive margin to investigate the effects of anchor firms on incumbent firm produc-

tivity and management practices. This paper presents a framework for understanding the impact of anchor firms on entrepreneurship while also capturing flows of labor out of incumbent firms. Finally, I discuss a setting where spillover effects are not driven by shocks directly to an incumbent firm's local industry, but instead to the supply chain.

The determinants of wages in the short-run are determined mostly by industry and location factors external to the firm (Lazear and Oyer, 2004). However, firms are tasked with responding to their external environment through reorganization. While the data does not allow me to determine whether firms are explicitly reorganizing by adding and dropping layers (Caliendo, Monte, and Rossi-Hansberg, 2015), firms exhibiting fastest wage growth after the anchor firm's arrival also shed the most workers who form spinoffs. Though these workers earn more relative to their industry, county, and industry-county pair, they are not necessarily the top wage earners in the firm.[14] This behavior of employee transitions in firms undergoing the most rapid post-anchor arrival wage growth is consistent with literature on firm reorganization (Caliendo et al. 2015) as well as compensation policies for and mobility of mid-level managers (Lazear and Shaw, 2007). Incumbent firms' efforts to reorganize in response to rising labor costs arising from the anchor firm's arrival may reflect improvement in management practices. Bloom et al. (2019) find evidence that increased density of economic activity in a county leads to improvements in structured management, perhaps reflecting reorganization and personnel management in surviving incumbent firms after the anchor firm arrives. Reorganization and structured management practices also would be reflected in measured labor productivity improvement in surviving incumbent firms found by Greenstone, Hornbeck, and Moretti (2010).

[14] I find evidence they may be above the firm average, but not in the top 3 or 5 wage earners.

2.2 Agglomeration

Agglomeration economies suggest spinoffs will collocate near their parent firms after a shock to the overall density of firms and labor markets. Dense labor markets facilitate better matching of workers to their employers particularly in industries relying on specialized labor (Duranton and Puga, 2004; Moretti, 2011). These thick labor markets improve sorting of workers into firms and occupations (Helsley and Strange, 1990) which enables firms and workers to appropriate returns to skill (Rotemberg and Saloner, 2000). If the introduction of an anchor firm increases density in local supply chain industries, then the composition of workers in supply chain startups will change. This selection effect of who enters entrepreneurship will right shift the distribution of startup quality and performance.[15]

Observational evidence shows that industries in the same supply chain tend to collocate (Ellison, Glaeser, and Kerr, 2010). One explanation as to why supply chains are spatially concentrated is that proximity to buyers and suppliers improves a firm's ability to monitor suppliers and tailor products to customers. This mechanism is most important in industries that are more knowledge intensive. For example, Alcacer and Delgado (2016) spatial clustering tends to be most pronounced for R&D facilities versus sales and production units. Even large and vertically integrated firms rely on external suppliers and contract with other firms to transport and ship goods (Atalay, Hortacsu, and Syverson, 2014). Some large firms strategically open establishments in clusters as a substitute for vertical integration (Helsley and Strange, 2006). For entrepreneurs, proximity to buyers and suppliers may be of particular importance. Glaeser and Kerr (2009) show the importance of thick local supply chain markets in supporting entrepreneurship.

[15] Combes et al. attribute half the spatial variation in firm productivity differences to selection of workers to firms.

2.3 Spinoffs

Another perspective on worker reallocation into young firms departs from traditional agglomeration mechanisms but instead relates to the literature on spinoffs. Spinoffs are empirically defined as instances where employees depart their employer to found a new firm that operates in the same or similar industry. Often, spinoffs are thought to be disruptive to their industry and potentially displace the incumbent employer. Why employees spinoff and which firms facility such transitions from wage-earning employment to firm founder has two main explanations. Some incumbent firms serve as incubators for workers with entrepreneurial ability and other incumbent firms suffer from organizational challenges that do not allow workers to appropriate returns to their inventiveness. I refer to this mechanism as the "bureaucratic channel."

Gompers, Lerner, and Scharfstein (2005) and Chatterji (2009) show that larger firms, often previously venture-backed firms, provide employees with knowledge essential to forming a business. Relevant knowledge for an entrepreneur may be non-technical in nature and include regulatory strategy, marketing, and connections to financiers and investors. In other specialized industries, technical expertise provided to a potential entrepreneur by a high quality employer is also particularly important. Franco and Filson (2006) show that high performing incumbents firms produce the most spin-outs. Interestingly, prior literature has not discussed whether incumbent firms face employee departures due to spinoffs because of shocks outside their industry.

The bureaucratic channel focuses on departures of top performing employees. Contracting frictions between these employees and upper management induce disagreements over which high-value projects to pursue and how to appropriate their returns. Spinoffs may arise because older firms are less able to add new product varieties in response to industry shocks (Franco and Filson, 2006; Klepper and Sleeper,

2005). Spinoffs through disagreements explain a number of industry clusters including Detroit automobiles (Klepper, 2007) and the Akron tire industry (Klepper and Buenstorf, 2009). A similar argument suggests that employee departures from incumbent firms into entrepreneurship arise because some employers are unable to properly appropriate returns to their employees' innovations (e.g., Anton and Yao, 1995; Chatterjee and Rossi-Hansberg, 2012).

3 Data

3.1 Anchor Firm Data

Site Selection is a corporate real estate publication with issues dating back to the late 1960s. The magazine variably publishes either quarterly or bi-monthly depending on year. Traditionally, the magazine included regular features titled "Location Reports", "Million Dollar Plants", "Blockbuster Deals" and "Top Deals" in which companies, generally Fortune 500 firms, disclosed information on the selection of sites for new facilities. The articles often, but not always, indicate the projected size of the opening facility, number of employees, and incentives/subsides offered by local economic development bodies. The announcements in the articles are sizeable economic events. They tend to involve openings of plants that are stated to eventually at least five hundred employees with an initial investment that often totals over a quarter of a billion dollars. Because of the size of the events, I refer to these new facilities as "large establishments" and their associated firms as "large firms."

I hand collected information on each announcement reported between 1990 and 2015 using historical microfiche archives, online editions, and searches on the magazine's website. I confirmed events and their locations through LexisNexis and Google searches. I assign company industry (NAICS) designations to the announcements us-

ing descriptions of the opening facility in the articles which often describe the product being manufactured or characteristics of the facility and/or company. I refer to these companies as "anchor firms."

Following other studies that have used these articles (e.g., Greenstone, Hornbeck, and Moretti 2010 and Bloom et al. 2019), I refer to these announcements as "cases" and the openings of these facilities as "plant openings." For each case, I document the year of the event, name of the firm, the county selected (referred to as *winners*), and other counties the firm considered at the end of its location selection process (which refer to as *finalists* and *runners-up* interchangeably). Most cases have one winner and one runner-up, but in select instances the article will mention multiple alternate sites the firm considered before its final decision. Most of the events involve manufacturing facilities, though R&D labs, telecommunications services, financial institutions, and corporate offices occasionally appear as well. Table 1 displays the distribution of industries by sector as well as the ten most frequently appearing NAICS-4 industries associated with the facilities openings in the sample.[16]

Table 1b shows the large size of firms represented in the magazine.

Figure 1 describes *Site Selection* magazine's geographic detail. The map shows total number of cases the magazine references for each state as having either a winning or runner-up county. All but six states are found in the hand-collected magazine sample. The map shows that the magazine over-represents states in the southeast, though states with the largest numbers of cases are distributed across the west coast, midwest, and mid-Atlantic.[17]

Table 2 displays results on comparing similarity between winners and runners-up

[16]Tables 1, 1b, and 2 do not describe specific locations and/or firms that may or may not be in the Census microdata. These tabulations are based solely on firms in the *Site Selection* magazine sample that are matched to Compustat.

[17]Figure 1 describes the data coming from *Site Selection*, but not the internal Census data that the magazine sample is matched to or the resulting matched dataset.

Table 1: Distribution of Announcements by Industry

A. Industry Sectors	Cases	B. 10 Most Common Announcement Industries	Cases
11. Agriculture	0	3361. Motor Vehicle Manufacturing	19
21. Mining	1	3364. Aerospace Product and Parts Manufacturing	14
22. Utilities	4	5417. Scientific Research and Development Services	12
23. Construction	0	3254. Pharmaceutical and Medicine Manufacturing	9
31-33. Manufacturing	134	3363. Motor Vehicle Parts Manufacturing	9
42. Wholesale Trade	11	3344. Semiconductor and Other Electronic Component Manufacturing	8
44-45. Retail Trade	15	5171. Wired Telecommunications Carriers	6
48-49. Transportation & Warehousing	12	5222. Nondepository Credit Intermediation	6
51. Information	12	3341. Computer and Peripheral Equipment Manufacturing	5
52. Finance & Insurance	22	5221. Depository Credit Intermediation	5
53. Real Estate	0		
54. Professional, Scientific, and Technical Services	12		
55. Management of Companies	4		
56. Administrative and Support, Waste Management	4		
61. Educational Services	0		
62. Health Care	2		
71. Arts, Entertainment, Recreation	1		
72. Accomodation & Food Services	5		
81. Other Services	1		

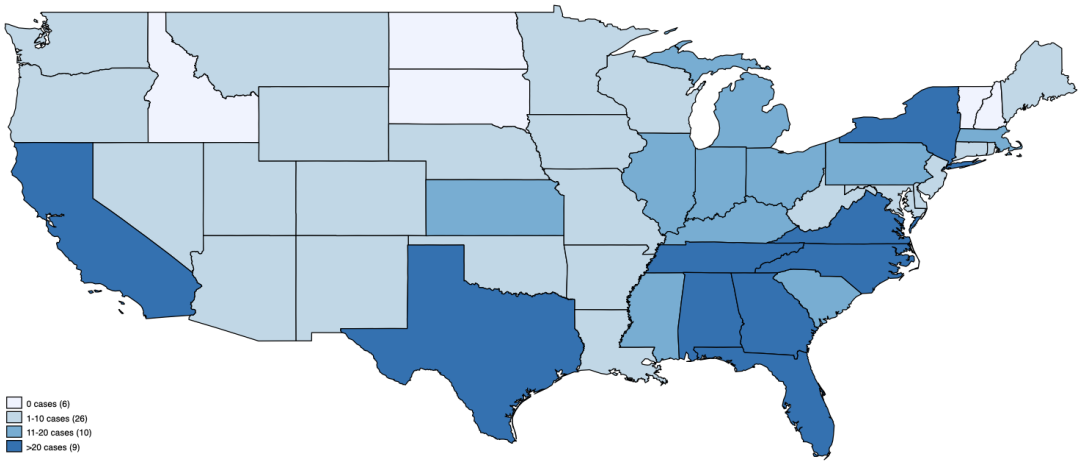
Notes: This table is not produced using Census microdata. Firms in this sample may or may not be in the Census microdata. The industry designations come from descriptions of the facility announcements in the magazine as well as Google searches on the facilities. This table shows that the anchor firm sample includes observations from a broad set of industries, though it over-represents the manufacturing sector. Panel A displays all private sector NAICS-2 industry sector designations and the number of *Site Selection* magazine large facilities openings announcement cases associated with each. Panel B shows the 10 most common industries appearing in the sample at the 4-digit NAICS definitions.

Table 1b: Firm Characteristics

	5th Percentile	Median	95th Percentile
Total Employees	899	37,300	362,063
Total Assets ($ Billions)	0.2898	15.4434	1097.19
Total Revenues ($ Billions)	0.2894	12.9983	1,128.915
Property, Plant, and Equipment ($ Billions)	0.0815	6.0944	828.997
R&D Expense ($ Billions)	0	0.7167	9.571
Share Foreign	0.3316		

Notes: This table is produced using only matches between all announcements in *Site Selection* magazine with complete information, Compustat, and S&P Capital IQ. Firms represented in this sample may or may not be in the matched Census sample. At the overall firm-level, the firms appearing in *Site Selection* magazine typically employ over 37,000 full-time employees and have $15 billion in assets. The median firm conducts nearly a billion dollars of R&D. Most firms represented are domicile in the United States, however one-third are foreign firms. Most foreign firms are Japanese. The tabulations are based off the year in which the firm made the announcement for each case. In cases where Compustat does not provide financial information, the tabulations use the year nearest to the announcement.

Figure 1: Geography of *Site Selection* Announcements



counties on dimensions commonly thought to be correlated to entrepreneurship. On all dimensions, the set of contest counties (those that are either a winner or runners-up) differ substantially from the average county in the U.S. In my subsequent analysis, I will use Table 2 and similar measures pertaining to human capital and employment in the supply chain to support the view that existing local human capital is an important driver in new venture formation and success in the supply chain.[18]

Unobserved differences in firm quality composition over the years immediately preceding the announcement potentially confound a difference-in-differences estimation. Firm quality in a county can drive both selection of an anchor firm into a location as well as the differences in the strategic response of firms in an area. To account for this, I check for parallel trends in the outcomes of interest as described in Section 6.1. In addition, a long empirical literature literature in firm dynamics discuss firm age and firm size distributions as important measurements of economic dynamism. Using Kolmogorov-Smirnov tests for the equality of firm age and firm size distributions in

[18]Namely, I identify that anchor firms target counties that disproportionately have thicker input-output markets. This could indicate higher levels of human capital or resources in these counties that support firms in these industries.

Table 2: Similarity of Winning and Runners-up Counties

	Winners	Runners-up	Difference	All counties
Number of counties:	168	205		3,182
Demographics				
Total population	917,390	1,273,125	355,735***	454,197
Total employment	230,910	236,700	-5,790	32,427
Non-white share	29.44%	30.29%	-0.85%	22.10%
Immigrant share	11.19%	12.41%	-1.22%	8.91%
Share older than 65	11.10%	11.35%	-0.25%	12.31%
% High school diploma only	23.87%	23.50%	0.37%	27.93%
% at least Bachelor's degree	32.57%	32.87%	-0.30%	27.28%
Economic				
Prime aged male joblessness	13.96%	14.88%	-0.92%**	14.46%
Self-employment rate	10.28%	10.30%	-0.02%	10.71%
Avg. income	$46,971	$47,610	-$639	$43,105
Mfg. share of employment	14.44%	13.81%	0.63%	11.42%
Single family home value (per sq. ft.)	$88.72	$91.71	-$2.99	$67.77
Community bank deposit share	13.65%	10.72%	2.93%*	8.05%
Announcement industry				
Employment	9,714	9,702	12	474
Employment share of county	3.08%	2.26%	0.81%**	1.38%
Establishment size	72	71	0.91	14

Notes: * 10%, ** 5%, *** 1%. The "All Counties" group differs from runners-up and winner counties at the 1% level for all categories except prime aged male joblessness and self-employment rate. All data reflect values from the mid-point of the sample period in the year 2000. Total population, non-white share, immigrant share, share older than 65, % high school diploma only, % at least bachelor's degree, prime aged male joblessness, self-employment rate and average income are obtained from the Census 2000 5% PUMS sample. Total population includes all individuals living in the county in the 2000 Census. Non-white share, immigrant share, and share $>$ 65 years of age are tabulated as the number of individuals in those categories divided by total population. The remaining PUMS variables are tabulated for prime aged males only. Prime aged males are males aged 25 to 55. Home values per square feet are tabulated using data from Zillow for March 2000. The sample matched to Zillow includes 134 winning counties, 169 runners-up and 1,499 in the 'all counties' sample. Community banks are defined as deposit taking institutions with assets less than $1 billion and either federally chartered and regulated by the Office of Thrift Supervision or state chartered and regulated by the FDIC. Total employment, mfg. share of employment, and all "Case-industry" variables are calculated from County Business Patterns for the year 2000. Employment in County Business Patterns reflect full-time employees on payroll on Tuesday the week of March 12 by county-industry. The tabulations are calculated at the 3-digit NAICS level with total employment reflecting the employment total across all 3-digit NAICS industries in the county. Mfg. share of employment is the percentage of employment between NAICS 310 and 339. The announcement industry variables are defined for the industries of firms in the Site Selection magazine sample. Employment shows the average employment levels in the industry grouped by county, employment share of county is the average employment share of the county employed in the industry, establishment size is the county-level employment over establishment count, and share of industry in county denotes the share of national industry employment in the county. This table is not produced using any administrative Census microdata nor does it indicate which cases may be matched into the Census.

the industries of interest as well as their input-output linked industries (Section 4.1), I fail to reject the null hypothesis of their equality at conventional levels of statistical significance. However imperfect these measures may be in summarizing the similarity of existing firms and workers in a county that may drive selection into the treatment as well as post-treatment outcomes, these checks further corroborate existing literature's usage of firm site searches in understanding local labor market dynamics and

firm or worker outcomes.

3.2 Longitudinal Business Database

I obtain employment, age, and location information U.S. business establishments from the restricted-use administrative Longitudinal Business Database (LBD) covering the period 1990-2015. The Census Bureau compiles a register of businesses using annual information on U.S. employer establishments from the Social Security Administration's Form SS-4, IRS Master Business File, IRS Form 941 and IRS Form 944.[19] The LBD files are at the annual level, but include time-invariant establishment identifiers allowing researchers to construct a panel of all U.S. business establishments. The LBD covers the universe of all non-farm, non-education/religious employer establishments. Establishments are defined as physical locations of business activity. The industry designations are provided at the 6-digit NAICS level. One particularly attractive feature of the LBD is that industries are assigned at the establishment rather than firm level. Industry assignments reflect the primary economic activity occurring at the physical location.

I use the LBD to measure startup activity. Though the LBD includes firm identifiers, researchers cannot define a startup as the first occurrence of the firm identifier. Transitions from being a single-establishment firm to a multi-unit firm often results in a reassignment of the Census firm identifier as well as the federal Employer Identification Number (EIN). To ensure firms are not mistakenly classified as startup from mechanical reassignments of firm identifiers, I do the following: Drop establishments that never report positive employment, label a firm as a startup only if all establishments associated with the parent firm appear for the first time, and calculate the

[19] Jarmin and Miranda (2002) provide technical documentation for the Longitudinal Business Database and its construction.

age of a firm as the age of the oldest establishment when firm identifier first appears and then allow the firm to age naturally each year regardless of changes in its establishment composition. This procedure mimics that of Decker et al. (2015). The key variables of interest tabulated from the Longitudinal Business Database include county-industry firm entry rates, firm growth rate, and firm survival.

3.3 Longitudinal Employer Household Dynamics

The Longitudinal Employer Household Dynamics (LEHD) database allows researchers to construct work histories for all workers employed in a participating state.[20] Observations are at the worker-firm-year-quarter level. Wage and employer information is obtained through state unemployment insurance offices. The LEHD is a complex network of files detailing quarterly wages for workers matched to a business reporting unit. The LEHD also provides geographic information as well as a limited set of demographic variables such as birth country, education, gender, age, and race. A limiting feature of the LEHD in studies over a long time horizon is that states enter the program in different years. Only 11 states participated in the program when it first began in 1992 with many of the current 31 states entering in the late 1990s through 2002. Researchers are granted access to states on a case-by-case basis with many only at the discretion of individual states themselves.

The LEHD is a complex network of files for each state. Part of the complexity of the LEHD is driven by the following: Firms may hold multiple federal EINs with each EIN representing a different tax reporting entity of the firm. Because the LEHD is organized at the state level with data provided through the ES-201 program through state unemployment insurance offices, employers are identified by state tax

[20] Vilhuber and McKinney (2014) provide an extensive documentation of the LEHD infrastructure files and source data.

ids (SEINs). Just as firms have multiple EINs, they may also have multiple SEINs assigned to them by each state in which they operate. Employees are assigned unique personal identifiers which follow the workers throughout the panel and can be matched to their SEIN employer at the tax reporting level (SEINUNIT). Unlike establishment identifiers in other Census products including the LBD, SEINUNITs are not defined by physical locations and are instead tax reporting units within the SEIN. However, the Census does provide a concordance file allowing matching of SEINs to EINs. The difference in establishment definition prevents the perfect mapping of physical establishments in the LBD to a SEINUNIT in the LEHD. Similar to other Census Bureau working papers such as Tate and Yang (2013) and Babina (2017), I use the combination of EIN-county, and entry date when possible, to obtain approximate employment count, industry, and age of the physical location of worker's job in the LEHD.

I use the Longitudinal Employer Household Dynamics data to construct work histories of individuals who ever start a firm. In the main employee-employer sample, I retain worker-firm-year observations for which average quarterly earnings in quarters worked at the firm is at above a full-time minimum wage salary of $2,678. To identify startups, I use the Longitudinal Business Database to obtain a list of firm-year observations for startup firms. I match these firm-years into the LEHD and the match yields a file of all employees in young firms. Empirically defining an entrepreneur in microdata is a challenge. The challenge arises because founders often are not compensated through income reported on W-2 filings from which LEHD earnings show. Second, Kerr and Kerr (2016) use Mark Zuckerberg's Facebook and Elon Musk's Tesla as examples to note that firms can be considered startups and individuals entrepreneurs long after they initially appear in the microdata. I follow the literature and label founders (also referred to as entrepreneurs in this proposal) to be the top three wage earners in years up to two years after the firm first appears

in the LEHD. I then use this list of entrepreneurs to create a panel of their employer and wage histories. This panel allows me to construct variables measuring employee transitions across industries and firms, migrate geographically (in terms of county of employment), and the likelihood of starting a new firm given industry experience.

I use the LEHD to construct four separate, but related samples. Two samples are used to compare winning counties to runners-up counties and the remaining two compare the winning county relative to itself in a pre-period. Sample 1 compares winning and finalist counties at the firm-level. This sample will be used to identify which firms produce the most entrepreneurs. Sample 2 compares contest counties at the entrepreneur-level to capture human capital specific factors in starting a supply chain firm. The last two samples are analogous but focused only on capturing changes in the composition of entrepreneurs pre- and post-anchor firm opening within the winning county itself. I construct additional samples to ensure the winning and runners-up counties do not differ from each other in the outcomes of interest prior to the opening event.[21]

4 Empirical Strategy

4.1 Defining Economic Proximity

One data limitation is that I do not observe firms' business transactions. This means I cannot directly ascertain whether an entrant is selling to or buying from the anchor firm's new facility. Instead I consider economically proximate firms to be those in industries that disproportionately purchase inputs from and sell outputs to the anchor

[21]The main specification take the set of firms and individuals in year $t-2$ and estimate entrepreneurial outcomes within the first 7 years of the anchor firm's arrival. The pre-analysis sample takes the set of firms and individuals in year $t-8$ and estimates whether winning and finalist counties differ in entrepreneurial outcomes through year $t-2$.

firm's industry. I refer to these industries as the anchor firm's "supply chain." I define the supply chain at the 4-digit NAICS level for each case in the *Site Selection* facility openings sample using the Bureau of Economic Analysis Use of Commodities tables. The supply chain refers to firms in the industry of anchor firm, firms in industries upstream to the anchor firm's industry, and firms in industries downstream to the anchor firm's industry.

The empirical definitions of upstream and downstream come from tabulations of industry-level output and input dependency. An industry's output dependency, $Output_{i \to j}$, is the share of industry i's outputs sold to industry j. I let the anchor firm's industry be industry i and label downstream ("buyer") industries for each industry i to be the leading five industry j's corresponding to the largest values of $Output_{i \to j}$. Table 4 lists the top 10 announcement industry-buyer industry pairs in the sample. Similarly, $Input_{i \leftarrow j}$ designates the share of industry i's inputs from industry j (Table 3). The five industry j's corresponding to the largest values of $Input_{i \leftarrow j}$ are upstream ("supplier") industries.

Table 3: Leading Upstream Industries

Announcement industry	Input provider	Share of inputs received from input producer
3241. Petroleum and Coal Products Manufacturing	2110. Oil and gas extraction	0.89
5250. Funds, Trusts, and Other Financial Vehicles	5239. Other Financial Investment Activities	0.76
3314. Nonferrous Metal (except Aluminum) Production and Processing	3314. Nonferrous Metal (except Aluminum) Production and Processing	0.74
3252. Resin, Synthetic Rubber, and Artificial Synthetic Fibers and Filaments Manufacturing	3251. Basic Chemical Manufacturing	0.69
5242. Agencies, Brokerages, and Other Insurance Related Activities	5242. Agencies, Brokerages, and Other Insurance Related Activities	0.68
3361. Motor Vehicle Manufacturing	3363. Motor Vehicle Parts Manufacturing	0.64
3222. Converted Paper Product Manufacturing	3221. Pulp, Paper, and Paperboard Mills	0.62
3251. Basic Chemical Manufacturing	3251. Basic Chemical Manufacturing	0.61
5241. Insurance Carriers	5418. Advertising, Public Relations, and Related Services	0.61
3116. Animal Slaughtering and Processing	1123. Poultry and egg production	0.60

Notes: Upstream industries are industries that serve as suppliers to the anchor firm's industry ("Announcement Industry"). Data is tabulated using the Bureau of Economic Analysis's "Use of Commodities" tables for 2007 and 2012. The rows in this table should be read as, for example, "52% of inputs used by the computer and peripheral equipment manufacturing industry is purchased from the semiconductor and other electronic components manufacturing industry." The table displays the ten industry pairs with the greatest buyer relationships and only includes buyer industries in the plant openings sample.

Table 4: Leading Downstream Industries

Announcement industry	Purchasing industry	Share of output sold to purchasing industry
3366. Ship and Boat Building	4830. Water Transportation	0.79
3361. Motor Vehicle Manufacturing	3362. Motor Vehicle Body and Trailer Manufacturing	0.75
3252. Resin, Synthetic Rubber, and Artificial Synthetic Fibers and Filaments Manufacturing	3261. Plastics Product Manufacturing	0.66
5239. Other Financial Investment Activities	5250. Funds, Trusts, and Other Financial Vehicles	0.63
3363. Motor Vehicle Parts Manufacturing	3361. Motor Vehicle Manufacturing	0.58
3362. Motor Vehicle Body and Trailer Manufacturing	3361. Motor Vehicle Manufacturing	0.55
3371. Household and Institutional Furniture and Kitchen Cabinet Manufacturing	2334. Building, Developing, and General Contracting	0.55
4231. Motor Vehicle and Motor Vehicle Parts and Supplies Merchant Wholesalers	3361. Motor Vehicle Manufacturing	0.52
3259. Other Chemical Product and Preparation Manufacturing	3231. Printing and Related Support Activities	0.50
5250. Funds, Trusts, and Other Financial Vehicles	5242. Agencies, Brokerages, and Other Insurance Related Activities	0.47

Notes: Downstream industries purchase inputs from the anchor firm's industry ("Announcement Industry"). Data is tabulated using the Bureau of Economic Analysis's "Use of Commodities" tables for 2007 and 2012. The table should be read as "66% of output produced by firms in the resin, synthetic rubber, and artificial synthetic fibers manufacturing industry is purchased by plastics products manufacturing firms." The table displays the ten industry pairs with the greatest buyer relationship. The focal industries on the left most column are restricted to the large facilities openings industries.

4.2 Difference-in-Differences Estimation

I use the introduction of a large facility of a large firm as a shock to both the local return to as well as the opportunity cost becoming an entrepreneur. I use the revealed preferences of large firms in the *Site Selection* announcements as a quasi-experiment to capture sudden opportunities that arise for individuals with specialized human capitals. These opportunities are represented as sudden changes in the proximity of a large potential buyer or supplier for an entrepreneur. The underlining assumption is that conditional on observed factors that lead an anchor firm to open a new facility in one county over another are orthogonal to unobserved characteristics that predict new firm formation and spinoffs by incumbent firms.

Table 2 provides descriptive statistics and test for equality between counties on a number of factors commonly thought to be related to entrepreneurship, the winners and runners-up counties are statistically similar. The tabulations are constructed by taking the averages of counties in each category: Winners, runners-up, and all. The set of factors listed are a small subset of the features that drive selection into entrepreneurship and regional startup rates. The table displays figures for the year

2000 since that year is near the center of period over which openings events were announced and for whom data is consistently available across all publicly available demographic and economic data sets. The variables include some of the most important traits including immigration status, education, size of the manufacturing base (Chinitz 1961), home values[22], labor market conditions (Fairlie 2013), presence of community banks, and establishment sizes (Glaeser, Kerr, and Kerr 2015).[23] I also test the overall establishment size distribution across each supply chain component. I find that winners and runners-up counties do not differ from each other in the distribution of establishments by firm size.[24]

Assuming the winner and runners-up counties within each site selection case are similar on dimensions related to entrepreneurship conditional on observable county-specific characteristics, I estimate a difference-in-differences model. The general form of the equation I estimate is:

$$\begin{aligned} y_{fijct} =& \beta_0 + \beta_1 \times \textit{Pre-trend}_{jt} + \beta_2 \times \textit{Post}_{jt} + \beta_3 \times \textit{Win}_{jct} + \\ & \beta_4 \times \textit{Win}_{jct} \times \textit{Post}_{jt} + \beta_5 \times \textit{Win}_{jct} \times \textit{Post-trend}_{jt} + \\ & \alpha_j + \lambda_{it} + \varepsilon_{fijt}, \end{aligned} \tag{1}$$

where f denotes a firm in industry i, case j, and county c at time t, and y_{fijct} indicates the dependent variable of interest as described in the variables section. β_4 is the difference-in-differences estimator of interest. λ_{it} are industry-year fixed effects

[22]However, some studies including Hurst and Lusardi (2004) and Kerr, Kerr, and Nanda (2015) find evidence that wealth constraints and housing collateral are only modest barriers to entry.

[23]The data on demographic traits as well as prime aged male joblessness are taken from the 2000 Census Public Use Microdata Series which has more accurate population estimates for larger counties. Only nearly 600 of the 3,100 U.S. counties are in the file. However, all other traits are taken from County Business Patterns, Zillow, and the FDIC Summary of Deposits which have more complete coverage.

[24]I test whether winners and runners-up counties have the same number of establishments by size groupings.

to control for secular trends that vary by industry and ω_c capture county fixed effects. The case fixed effects, α_j, ensure the difference-in-differences estimators are identified off comparisons between the winner and runner-up within the specific announcement event. Reported regression coefficients are restricted to the set of startups in winners and finalist counties, but in robustness checks I use the set of all counties to enhance identification of the industry-year fixed effects. I consider the pre-period to be up through five years before the announcement article and the post-period to run through 7 years afterwards.[25]

To capture the impacts of founder background and firm characteristics I take the set of workers in year $t-2$ relative to the facilities opening and compare the number of employee departures for incumbent firms in winning and runners-up counties between t and $t+7$. I use the LEHD to observe employee transitions across firms and identify startup founders. Because the state-level LEHD uses a slightly different definition of a business than IRS-based LBD, I define incumbent firms at the SEIN-county level. I first take the set of all employees in each SEIN-county in year $t-2$ and use as the outcome variable the number of employees who depart the firm in the post period to start a firm in the supply chain. In founder-based regressions, my outcome variable is instead defined at the worker-level and I use a linear probability model to estimate the likelihood an individual starts a firm in the supply chain conditional on departing an incumbent firm to become an entrepreneur.

The baseline regression follows a simple difference-in-differences for the number of employee departures:

[25]The case fixed effects are dummy variables for the case number associated with the firm-county observation interacted with a dummy variable if the observation is between five years prior to and seven years post-announcement.

$$y_{fijct} = \beta_0 + \beta_1 \times Post_{jt} + \beta_2 \times Win_{jct} + \beta_4 \times Win_{jct} \times Post_{jt} + \beta_5 \times X_{fj} + \alpha_j + \lambda_{it} + \varepsilon_{fijt}, \quad (2)$$

Here, the firm f is at the SEIN-county level and X_{fj} is a vector of firm controls including total employment and age. To capture the change in likelihood that an entrepreneur selects into the supply chain, I define the outcome as y_{pijct} which takes on the value of 1 if the entrepreneur selects into a supply chain industry. These entrepreneur-level regressions control for race, sex, education, age, and age squared.

To estimate the variation across the supply chain by experience within specific components of the supply chain, I add industry interactions to Equation 2:

$$\begin{aligned} y_{fijct} = \beta_0 + \beta_1 \times Win_{fjc} + \beta_2 \times Upstream_{fijc} + \beta_3 \times Anchor_{fijc} + \beta_4 \times Downstream_{fijc} + \beta_5 \times Upstream_{fijc} \times Win_{fjc} + \\ \beta_6 \times Anchor_{fijc} \times Win_{fjc} + \beta_7 \times Downstream_{fijc} \times Win_{fjc} + \\ \beta_8 \times X_{fjt} + \alpha_j + \lambda_{it} + \varepsilon_{fijt}, \end{aligned} \quad (3)$$

I make the necessary substitutions of y_{pijct} in the outcome and X_{fjt} as a control for entrepreneur-level regression.[26] The coefficients of interest are β_5, β_6, and β_7.

To estimate the role of founder characteristics how firm and founder characteristics affect spinoffs as well as selection of entrepreneurs into supply chain industries, I run analogous regressions to Equation 3 by adding interactions with C_fjc, the char-

[26] Even though these regressions fix the firm (or eventual entrepreneur) in year $t-2$, the treatment of the facilities opening occurs anytime between 1995 and 2008 which is why the secular industry trend λ_{it} remains as a fully identified fixed effect.

acteristic of interest:

$$
\begin{aligned}
y_{fijct} = \beta_0 + \beta_1 \times Win_{fjc} + \beta_2 \times Win_{fjc} \times C_{fjc} + \\
\beta_3 \times Upstream_{fijc} + \beta_4 \times Anchor_{fijc} + \beta_5 \times Downstream_{fijc} + \\
\beta_6 \times Upstream_{fijc} \times Win_{fjc} + \beta_7 \times Anchor_{fijc} \times Win_{fjc} + \\
\beta_8 \times Downstream_{fijc} \times Win_{fjc} + \beta_9 \times Upstream_{fijc} \times Win_{fjc} \times C_{fjc} + \\
\beta_{10} \times Anchor_{fijc} \times Win_{fjc} \times C_{fjc} + \beta_{11} \times Downstream_{fijc} \times Win_{fjc} \times C_{fjc} + \\
\beta_{12} \times X_{fj}\alpha_j + \lambda_{it} + \varepsilon_{fijt},
\end{aligned}
\tag{4}
$$

where β_2, β_9, β_{10}, and β_{11} are the coefficients of interest. The excluded group from the industry designations are the "all other industries" category. This means that coefficients are interpreted as relative to a supply chain founder from an industry unrelated to the focal firm's defined supply chain. I interpret the β_3's to suggest the relative propensity of an entrepreneurial individual selecting into the announcement firm facility's supply chain.

To lend credibility that the opening of the anchor firm drives variation in the post period, I repeat all specification using only the pre-period. I compare outcomes for businesses and individuals in winners and runners-up counties between 8 and 2 years prior to the facilities openings. Overall, I find no evidence that the counties differ in the number of spinoffs incumbent firms yield or the likelihood of entrepreneurs to select into the industries of the anchor firm that eventually arrives. Online Appendix B outlines the robustness checks on spinoff and entrepreneur-level specifications.[27]

[27] An additional check is to measure the impact of effects by size of the arriving anchor firm.

5 Results

5.1 Economic Proximity and Internal Employment Growth

I first document changes in firm entry in the supply chain after the arrival of an anchor firm at the county-industry level.[28] I measure the entry rate of new businesses in a county as the share of establishments in the county-industry attributable to new startup firms. Following the literature, I consider startups to be firms aged less than three years.[29]

The sample consists of all firms in the contest county for each announcement event's associated supply chain. The supply chain is defined independently for each individual announcement case. Therefore, the industries included in case 1 are different than those in case 2. The number of new establishments of firms aged less than three and employment in these establishments are collapsed at the county-announcement event-industry level. This implies that for each contest county within a case there are annual observations for up to 5 supplier industries, 1 anchor industry, and 5 buyer industries. Industries are defined at the 4-digit NAICS level. The coefficients in Tables 5-8 represent the average effect the anchor firm has on entrepreneurship at the county-industry pair level. Column 1 of Table 5 shows an increase of about 23 new establishments in each industry-county pair of all supply chain industries. This represents an increase in county-industry entry rate of approximately 0.75%, or a 10% increase in the entry rate from the county-industry average. Column 2 shows that the impact on employment amounts to 429 new jobs added in county-industry pairs associated with the supply chain. In total, this amounts to over 120 total startup

[28] The null results showing no change in startup entry and startup employment have not yet been disclosed.

[29] I use firms aged less than three in part due to noise in microdata related to spurious entry in the initial year that are often populated by 0's for employment, missing payroll, or a 0 for either employment or payroll but not both.

establishments and 2,300 jobs in startups across all unique supply chain NAICS-4 industries represented in an average case.

Table 5: Value chain county-industry entry and startup growth

	(1)	(2)	(3)	(4)
	Establishments of new firms	Employment from new firms	5-year growth rate	5-year failure probability
Constant	7.389***	3.301***	-0.3378***	0.166***
	(0.114)	(0.123)	(0.0160)	(0.024)
Win	-0.8781	23.62	-0.0410	0.01
	(10.24)	(125.2)	(0.0346)	(0.76)
Post	-6.607*	-489.1***	-0.0190	0.01
	(3.446)	(120.3)	(0.0182)	(1.60)
Win * Post	22.05*	429.6**	0.1225*	-0.0680*
	(11.98)	(185.1)	(0.0663)	(0.0340)
Win * Post-trend	-1.538*	-28.82**	-0.0087*	0.005*
	(0.892)	(13.98)	(0.0046)	(0.003)
Pre-trend	-0.573**	-34.93***	-0.0022	0.00
	(0.236)	(10.83)	(0.0020)	(1.50)
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Effect in t+5	12.82*	256.7**		
Adj. R-Sq.	0.7552	0.6032	0.0988	0.1330
N	91500	91500	215000	215000

Notes: * 0.1 ** 0.05 *** 0.01. The anchor firm's arrival induces the formation of supply chain startups that add 430 new jobs to the county. These startups grow 12% faster in employment over the first five years and fail 7% less. This table displays output from a difference-in-differences regression estimating the impact of a large facilities opening on the outcome indicated for all industries in the supply chain. The unit of observation is at the county-industry-year level. Industries are defined at the 4-digit NAICS level. Industries are classified as being in the same industry as the opening announcement, as a supplier firm (upstream), or buyer firm (downstream). Supplier and buyer are defined as the top 5 non-zero industry of goods & services flows by the Bureau of Economic Analysis Use of Commodities tables. The outcome variable in Column 1 is the number of new startups at the county-industry level. Column 2 shows the total employment attributable to these new businesses. Column 3 shows the 5-year growth rate of firms after their first year of appearing in the Longitudinal Business Database. The growth rate is calculated using the Davis, Haltiwanger, and Schuh (1998) firm growth calculation aggregated by establishment-level microdata. The measure is symmetric on the interval [-2, 2], where -2 indicates firm exit and 2 indicating firm entry into the industry. Column 4 displays the impact of a facilities opening on the likelihood the startup fails within five years. Column 4 uses a binary variable of 1 indicating the startup fails within five years as the outcome. "Effect in t+5" shows the number of new establishments from new firms (Column 1) and new job from these establishments (Column 2) in year t+5. These values are not shown for Columns 3 and 4, though the results are qualitatively similar to the difference-in-differences estimator. All regressions are weighted by total county-industry employment. Census rounding rules limit the number of significant digits reported to 4. The number of decimal places reported in each column use the minimum number after Census rounding. Standard errors are clustered by county.

Tables 6, 7, and 8 show that the aggregate magnitude is largely driven by upstream entry followed by downstream entry. Entry does not occur in the anchor firm's industry. First restricting to county-industry pairs defined as upstream within each announcement event in Table 6, Column 1's estimate of the difference-in-difference estimator shows a treatment effect amounting to an increase in 23 new startup establishments associated with 453 (Column 2) jobs at the county-industry level. This

increase amounts to about 15% of total upstream employment. Though currently undisclosed, much of this upstream entry is driven by startup formation in the first three years of the anchor firm's arrival.

Table 6: Upstream county-industry entry and startup growth

	(1)	(2)	(3)	(4)
	Establishments of new firms	Employment from new firms	5-year growth rate	5-year failure probability
Constant	76.34***	1322***	-0.3482***	0.167***
	(4.55)	(91)	(0.0217)	(0.017)
Win	-6.72	-190.4*	-0.0678	0.03
	(10.82)	(101.6)	(0.0535)	(1.12)
Post	-10.36	-530.4***	-0.0588*	0.0425**
	(6.89)	(197.0)	(0.0317)	(0.0220)
Win * Post	23.50***	453.1*	0.1888*	-0.104*
	(8.06)	(254.1)	(0.1094)	(0.063)
Win * Post-trend	-1.836***	-27.10*	-0.0135*	0.0075*
	(0.682)	(16.16)	(0.0076)	(0.0045)
Pre-trend	-1.322**	-47.20***	-0.0086**	0.0048**
	(0.611)	(15.24)	(0.0039)	(0.0023)
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Effect in t+5	12.49***	290.5*		
Adj. R-Sq.	0.7259	0.7717	0.1312	0.1660
N	32000	32000	88500	88500

Notes: * 0.1 ** 0.05 *** 0.01. After the anchor firm's arrival, the number of upstream startup establishments increases by 23 and represent 450 new jobs. These startups grow 19% faster and fail 10% less. This table displays output from a difference-in-differences regression estimating the impact of a large facilities opening on the outcome indicated for all upstream industries in the supply chain. The unit of observation is at the county-industry-year level. Industries are defined at the 4-digit NAICS level. The outcome variable in Column 1 is the number of new startups at the county-industry level. Column 2 shows the total employment attributable to these new businesses. Column 3 shows the 5-year growth rate of firms after their first year of appearing in the Longitudinal Business Database. The growth rate is calculated using the Davis, Haltiwanger, and Schuh (1998) firm growth calculation aggregated by establishment-level microdata. The measure is symmetric on the interval [-2, 2], where -2 indicates firm exit and 2 indicating firm entry into the industry. Column 4 displays the impact of a facilties opening on the likelihood the startup fails within five years. Column 4 uses a binary variable of 1 indicating the startup fails within five years as the outcome. "Effect in t+5" shows the number of new establishments from new firms (Column 1) and new job from these establishments (Column 2) in year t+5. These values are not shown for Columns 3 and 4, though the results are qualitatively similar to the difference-in-differences estimator. All regressions are weighted by total county-industry employment. Census rounding rules limit the number of significant digits reported to 4. The number of decimal places reported in each column use the minimum number after Census rounding. Standard errors are clustered by county.

Table 7 shows no increased startup activity in the anchor firm's own industry.

Table 7: Anchor firm county-industry entry and startup growth

	(1)	(2)	(3)	(4)
	Establishments of new firms	Employment from new firms	5-year growth rate	5-year failure probability
Constant	25.41***	641.7***	-0.4532***	0.186***
	(1.32)	(102.2)	(0.0793)	(0.071)
Win	-8.911***	-290.5**	0.0571	-0.0197
	(2.927)	(141.2)	(0.0633)	(0.0660)
Post	0.345	-23.5	0.1365	-0.0123
	(1.597)	(187.2)	(0.1458)	(0.0290)
Win * Post	6.134	16.4	-0.2488	0.0136
	(4.993)	(360.9)	(0.1612)	(0.1900)
Win * Post-trend	-0.2924	3.76	0.0119	-0.0015
	(0.3594)	(28.56)	(0.0123)	(0.2500)
Pre-trend	0.0803	104.7	0.0181	-0.0008
	(0.1791)	(69.3)	(0.0165)	(0.1700)
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Effect in t+5	4.380	38.98		
Adj. R-Sq.	0.9035	0.4986	0.1141	0.1540
N	12000	12000	10500	10500

Notes: * 0.1 ** 0.05 *** 0.01. Startup formation and performance in the anchor firm's industry are unaffected by the facilities opening event. This table displays output from a difference-in-differences regression estimating the impact of a large facilities opening on the outcome indicated for the anchor firm's industry. The unit of observation is at the county-industry-year level. Industries are defined at the 4-digit NAICS level. The outcome variable in Column 1 is the number of new startups at the county-industry level. Column 2 shows the total employment attributable to these new businesses. Column 3 shows the 5-year growth rate of firms after their first year of appearing in the Longitudinal Business Database. The growth rate is calculated using the Davis, Haltiwanger, and Schuh (1996) firm growth calculation aggregated by establishment-level microdata. The measure is symmetric on the interval [-2, 2], where -2 indicates firm exit and 2 indicating firm entry into the industry. Column 4 displays the impact of a facilities opening on the likelihood the startup fails within five years. Column 4 uses a binary variable of 1 indicating the startup fails within five years as the outcome. "Effect in t+5" shows the number of new establishments from new firms (Column 1) and new job from these establishments (Column 2) in year t+5. These values are not shown for Columns 3 and 4, though the results are qualitatively similar to the difference-in-differences estimator. All regressions are weighted by total county-industry employment. Census rounding rules limit the number of significant digits reported to 4. The number of decimal places reported in each column use the minimum number after Census rounding. Standard errors are clustered by county.

Table 8 restricts to county-industries downstream to the anchor firm. A comparison of Columns 1 and 2 shows that while the number of downstream startups do not increase, those that do enter at much larger scale than before with over 260 new jobs in new startup establishments. The effects last at least 5 years after the anchor firm arrives. Tables 6 and 8 show that in the fifth year after the anchor firm arrives, 291 jobs in upstream startups and 175 jobs in downstream startups are still being created.

Table 8: Downstream county-industry entry and startup growth

	(1)	(2)	(3)	(4)
	Establishments of new firms	Employment from new firms	5-year growth rate	5-year failure probability
Constant	7.896***	739.8***	-0.3182***	0.159***
	(0.180)	(53.8)	(0.0220)	(0.062)
Win	-6.083	-56.41	-0.0286	0.0035
	(11.03)	(85.76)	(0.0416)	(0.2002)
Post	-8.969	-42.0	-0.0072	-0.0040
	(6.188)	(123.8)	(0.0240)	(0.4401)
Win * Post	25.79	260.6**	0.1511**	-0.0593*
	(18.63)	(121.1)	(0.0741)	(0.0362)
Win * Post-trend	-1.829	-14.30*	-0.0092*	0.004
	(1.145)	(7.27)	(0.0052)	(1.900)
Pre-trend	-0.9872**	-4.63	0.0006	-0.0004
	(0.4386)	(12.18)	(0.0026)	(0.3400)
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Effect in t+5	14.82	174.8**		
Adj. R-Sq.	0.7966	0.3737	0.0953	0.1300
N	28000	28000	115000	115000

Notes: * 0.1 ** 0.05 *** 0.01. After the anchor firm's arrival, downstream startups generate 260 more jobs, grow 15% faster, and fail 6% less. This table displays output from a difference-in-differences regression estimating the impact of a large facilities opening on the outcome indicated for all downstream industries in the supply chain. The unit of observation is at the county-industry-year level. Industries are defined at the 4-digit NAICS level. The outcome variable in Column 1 is the number of new startups at the county-industry level. Column 2 shows the total employment attributable to these new businesses. Column 3 shows the 5-year growth rate of firms after their first year of appearing in the Longitudinal Business Database. The growth rate is calculated using the Davis, Haltiwanger, and Schuh (1998) firm growth calculation aggregated by establishment-level microdata. The measure is symmetric on the interval [-2, 2], where -2 indicates firm exit and 2 indicating firm entry into the industry. Column 4 displays the impact of a facility's opening on the likelihood the startup fails within five years. Column 4 uses a binary variable of 1 indicating the startup fails within five years as the outcome. "Effect in t+5" shows the number of new establishments from new firms (Column 1) and new job from these establishments (Column 2) in year t+5. These values are not shown for Columns 3 and 4, though the results are qualitatively similar to the difference-in-differences estimator. All regressions are weighted by total county-industry employment. Census rounding rules limit the number of significant digits reported to 4. The number of decimal places reported in each column use the minimum number after Census rounding. Standard errors are clustered by county.

One possible effect of the anchor firm's arrival is that startup entry occurs *en masse* due to expectations on the ease of finding suppliers and customers or per-

ceived signalling about the quality of the local industry. If the arriving anchor firm leads to more entry driven by unfounded beliefs on the local market, these entrants will be of low quality and fail quickly. To measure startup growth, I use the conventional definition for calculating firm growth from establishment-level microdata (Davis, Haltiwanger, and Schuh, 1998):

$$g_{ft} = \sum_{k \in f} \gamma_{kt} \times \frac{\text{emp}_{k,t+5} - \text{emp}_{kt}}{0.5 * (\text{emp}_{kt} + \text{emp}_{kt+5})} \tag{5}$$

where γ_{kt} is establishment k's share of firm f's total average employment between year t and $t+5$. My focus is on understanding how much firms grow specifically in the supply chain. Therefore, I measure firm growth with respect to the total employment in the supply chain position. For example, Table 6 uses a startup's growth in all upstream industries as the outcome.[30] This outcome variable is used to estimate Equation 1 and is reported in Column 3 of Tables 5-8. Table5 demonstrates that the effect of the anchor firm's arrival in winning counties on startup's supply chain growth is 12.25%. Table 6 shows that the impact on upstream startup growth is 18.88%. Table 7 continues to suggest the lack of an effect within the anchor firm's industry itself. Table 8 shows the treatment effect of a 15.11% increase in employment growth.

I also test effects on the survival of young firms. To accurately capture startup exits, I follow existing common practice and only consider a startup an exit if all establishments underneath a firm identifiers in year t exit by year $t+5$. Formally, in

[30] A startup could operate in industry A and industry B, both of which are supplier industries. I combine firm-county employment for these two industries and calculate the firm's upstream growth. To identify industry-by-year fixed effects, I use the NAICS-4 of greatest employment (i.e., greatest employment between A and B).

Equation I use the following outcome:

$$Pr(\text{Firm exit}_{ft} = 1), \tag{6}$$

if all establishments exit by year $t + 5$. Column 4 of Tables 5-8 show the magnitude of the treatment effect of an anchor firm on failure probabilities. Table 5 shows a decrease of 6.8% across all supply chain industries. Table 6 estimates a decline of 10.4% for upstream industries. Tables 7 and 8 and show no change in failure and a decline of 5.9% in anchor and downstream industries, respectively.

For all regressions, I test for parallel trends. I use a dynamic difference-in-difference where the win dummy variable is interacting for a time dummy for all times of the case period. I find no significant results on any of the time dummies. Please refer to Online Appendix B for details on these additional checks.[31] In addition, I show that aggregate employment and establishment counts across the full firm age distribution in the county and supply chain do not increase.[32]

In summary, Tables 5-8 show firm entry, new venture growth, and survival improves in upstream and downstream industries. This set of results shows: 1) Aggregate employment and establishment counts across all firm ages are unchanged using the set of all industries as well as limiting to supply chain industries for each announcement case. The arrival of the anchor firm appears to only shift the composition of industry employment from old to young in supply chain industries. 2) Anchor firms induce entry and employment gains in startups in upstream and downstream firms. 3) Anchor firms lead to faster growth and survival of supply chain

[31] Forthcoming when robustness check disclosure is passed.

[32] These additional results are under disclosure review as well. However, the aggregate results may also be identified using County Business Patterns. Effects of the anchor firm on total employment and establishment counts are near 0 across all industries, supply chain industries only, upstream industries only, anchor industries only, and downstream industries only.

startups. Similar to works discussing the importance of local factor markets in supporting entrepreneurship or clustering of firms (e.g., Glaeser and Kerr 2009), these results suggest potential increases in the thickness of local supplier-buyer networks support entry and startup performance.

5.2 Spinoffs and Employee Departures

All analyses in Section 4.2 are at the level of incumbent firms and their employees who spin-out in winning and runners-up counties. I find descriptive evidence that anchor firms typically focus their final location candidates to counties that have greater employment specialization in industries upstream and downstream to them (Table 2 and Online Appendix B). I use this descriptive tendency as motivation for probing at whether the treatment effect found in Section 5.1 is specific to founders of high human capital as measured by industry experience and wages. If high quality firms incubate entrepreneurs (Gompers, Lerner, and Scharfstein, 2005; Chatterji, 2009), then proxies of quality such as wages should relate positively to the number of entrepreneurs they produce. If industry experience is essential then not only will high quality firms generate more entrepreneurs, the entrepreneurs will select into their employer's industry. Finally, I investigate heterogeneity within incumbent firms to test whether firms lose more individuals to spinoffs or individuals of higher ability or ranking within the firm. Namely, I focus on industry experience and ability using earnings as a proxy. I conduct the analysis from two perspectives. In each characteristic of interest I first investigate which firms generate the greatest number of spinoffs as measured by the number of individuals who leave the firm to start their own in the post-opening period. The second set of regressions under each heading estimate a linear probability model of industry selection of entrepreneurs conditional on the

individual eventually becoming an entrepreneur.[33]

5.2.1 Worker Transitions and Economic Proximity

I first ask whether firm entry is driven by spinoffs or a general transition of workers to entrepreneurship regardless of industry background. Operationally I define a spinoff as firms founded by employees in the same segment (upstream, anchor firm industry, or downstream) of the supply chain as their most recent employer. Following the sample construction outlined in Section 4.2, Table 9 estimates Equation 3 and displays the key coefficients of interest on the triple interaction between the winning county indicator, runners-up county dummy variable, and the supply chain category variable. The baseline category is grouped as all industries outside the supply chain which means the coefficients are interpreted as the number of employee departures into entrepreneurship relative to firms outside the supply chain.

Table 9 uses the total number of employee departures into any industry of entrepreneurship. Looking within a column, the rows indicate the relative number of employee departures into entrepreneurship of that category. Column 1 shows that firms in winning counties after the anchor firm's arrival do not lose more employees to entrepreneurship. Column 2 shows that relative to firms outside the supply chain, upstream incumbents in winning counties lose .17 more employees after the arrival of the anchor firm. Relative to the baseline this coefficient represents an increase of over 30% in employee departures. Put differently, this coefficient implies the anchor firm's arrival leads to 1.7 new spinoffs for every ten upstream firms in the county. Moving through Columns 2, 3, 4, Table 9 shows that firms disproportionately lose employ-

[33]The idea behind estimating industry selection only on the set of eventual entrepreneurs is that I aim to measure changes in the composition of startups after the anchor firm arrives. However, this does assume that individuals decide to select into entrepreneurship and then pinpoint the industry they enter instead of first estimating the latent distribution of would-be entrepreneurs who stay employed.

Table 9: Number of Employee Departures to Entrepreneurship by Firm Industry

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Win * Upstream	-0.0914	0.1682***	0.0305***	0.0271**
	(0.0873)	(0.0258)	(0.0115)	(0.0119)
Win * Anchor Industry	-0.0981	0.4103***	0.4960***	0.1365*
	(0.3831)	(0.0814)	(0.0793)	(0.0784)
Win * Downstream	0.0403	0.0745***	0.0167	0.3968***
	(0.0910)	(0.0197)	(0.0177)	(0.0432)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.6367	0.1149	0.0596	0.0838
N	1094000	1094000	1094000	1094000

Notes: * 0.1 ** 0.05 *** 0.01. Winning county incumbent firms do not lose more employees to entrepreneurship after the anchor firm's arrival, but the industries their employees select into shifts in favor of supply chain industries. This table shows the variation in a facilities opening on employee departures to entrepreneurship by firm-industry of employment. The outcome variables are the number of employees in a firm who leave within 7 years of the opening to found a new firm. Firm founders are defined as a top 3 wage earner in a firm younger than three years. All regressions regress the outcome variable on a dummy variable indicating whether the firm is in a winning or runners-up county, industry classification in the supply chain, and interaction between winning status and each industry group. The excluded industry category is "all other industries." Industry groupings are determined for each case: Firm-industries are coded differently in case 1 versus case 2 depending on the anchor firm's industry in each case. Column 1's outcome variable is the total number of employee departures to entrepreneurship regardless of industry (including non-supply chain), Column 2 shows the number departing to form an upstream firm to the anchor firm, 3 for anchor firm's industry, and 4 for downstream industry startups. Regressions are weighted by firm size and standard errors are clustered at the SEIN-county.

ees to entrepreneurship in their own segment of the supply chain. Upstream and downstream firms lose employees disproportionately to upstream and downstream entrepreneurship and anchor industry firms generate entrepreneurs across the supply chain. The introduction of an anchor firm to the county also induces employee departures into supply chain entrepreneurship by employees in incumbent anchor industries.[34]

Table 10 shows selection into supply chain entrepreneurship conditional on an individual eventually becoming an entrepreneur. All coefficients represent the change in likelihood relative to the counterfactual counties that an entrepreneur selects into the specified industry (column headings) given their industry of employment prior to the anchor firm's arrival. Moving downwards diagonally across columns, the table shows entrepreneurs become significantly more likely to start a new venture in the supply chain position of their prior employer. Entrepreneurs from upstream (anchor firm) industries are 8.7% (4.4%) more likely to start an upstream (anchor industry) firm. The strongest effect is found in entrepreneurs from downstream firms who become 17.2% more likely to have their startup in a downstream industry. Table 10 also shows having experience in an anchor industry only has a significant and positive impact of forming an anchor industry startup and corresponds to a strongly negative (-17.3%) impact on founding a downstream venture.

Taken together Tables 9 and 10 demonstrate that the anchor firm's arrival alters the industry composition of entrepreneurship. While firms do not lose more employees to entrepreneurship than before, those that they do lose are more likely to form spinoffs and operate in a similar industry. The change in composition of founder background is driven by the opening of a large facility in the supply chain of the

[34]The coefficients allow for computing the implied share of new startups by a county's employees that are spinoffs: 25% of upstream, 91% of anchor industry, and 71% of downstream.

Table 10: Likelihood of Entry Industry by Industry of Prior Employment

	(1)	(2)	(3)
	Pr(Upstream startup)	Pr(Anchor industry startup)	Pr(Downstream startup)
Win * Upstream	0.0871***	0.0083*	-0.0054
	(0.0144)	(0.0046)	(0.0054)
Win * Anchor Industry	-0.0126	0.0444**	-0.1730***
	(0.0142)	(0.0179)	(0.0227)
Win * Downstream	-0.0115	-0.0286***	0.1720***
	(0.0071)	(0.0070)	(0.0188)
Main Effects	Y	Y	Y
Worker Controls	Y	Y	Y
Industry x Year FE	Y	Y	Y
Case FE	Y	Y	Y
Adj. R-Sq.	0.1053	0.0589	0.1485
N	2124000	2124000	2124000

Notes: * 0.1 ** 0.05 *** 0.01. The anchor firm's arrival increases the likelihood that an entrepreneur in the winning county selects into supply chain entrepreneurship. This table shows the variation in a facilities opening on likelihood an individual departs to entrepreneurship by firm-industry of employment. The sample takes individuals who ever start a firm in the post-opening period. Each of the three regressions estimate whether the entrepreneur started a firm in the specified industry (columns) depending on industry of employment in the year prior to the announcement event (rows). The excluded industry category is "all other industries." Controls include age, age-squared, sex, race, education, and SEIN FEs. Regressions are unweighted and standard errors are clustered at the SEIN-county.

county's incumbent firms. Table 9 also shows that anchor industry incumbent firms in treated counties generate more entrepreneurs across supply chain industries, but Table 10 suggests the likelihood of starting a firm in the supply chain is unaffected for upstream startups and reduced for downstream startups. These two facts may suggest that the number of employee departures into supply chain entrepreneurship by anchor firm industries is driven by departures by a concentrated set of firms such that across all entrepreneurs (Table 10) the likelihood of an entrepreneur coming from an anchor industry is largely unaffected by being in a treated county.[35] This finding potentially complements both Greenstone et al. (2010) and Bloom et al. (2019) who discuss productivity and management practices spillovers onto incumbent firms in the anchor firm's industry.

[35]Subsequent analysis in Section 5.2.2 will show the characteristics of anchor industry firms who lose employees to entrepreneurship in the supply chain.

One concern with estimating Equation 3 is pre-existing differences in spinoffs and entrepreneurial selection into specific industry prior to the anchor firm's arrival.[36] To account for this, I run the same analyses using incumbent firms in year $t-8$ and check spinoffs over the subsequent years up until the anchor firm's arrival and find no differences between the counties. In Section 4.3 I further compare winning counties against themselves from a long pre-period to verify that the change in composition of winning counties specifically is driving the findings.

5.2.2 Firm Compensation and Wage Compression

Though the literature is personnel economics generally does not focus specifically on the relationship between compensation and the characteristics of firms employees may transition to, factors such as average firm wage and pay compression may be closely related to employee departures. Outside of economics, the management literature studies use firm wages relative to the industry as well as pay dispersion as measurements of employee performance and firm quality. Examples include Campbell et al. (2012) and Carnahan, Agarwal, and Campbell (2012) who discuss that high paying firms retain top earners, but those who do leave select into entrepreneurship. These top earners typically have the most industry expertise and found successful ventures (e.g., Franco and Filson 2006).

Operating under the assumption that a firm's wages relative to the industry are reflective of firm quality, I estimate Equation 4 using mean wages in the firm as the firm characteristic of interest. Firm mean wages are tabulated as the average quarterly earnings of workers retained in the sample (Section 4) by SEIN-county. The variable is standardized to mean 0 and unit standard deviation. The industry-by-year fixed effect implies the coefficients on the interactions are estimated within

[36] This concern arises because the comparisons are made between winning and runners-up county between year $t-2$ and year $t+7$ without an explicit control from pre-existing differences.

each incumbent firm's industry and show how compensation differentials relative to the industry influence spinoffs. Pay dispersion in the firm is tabulated as the variance in average worker wages in the firm. This variable is similarly transformed to mean 0 and unit standard deviation.

The main wage interaction with the winning county dummy variable in Column 1 of Table 11 shows that firms with higher than industry average wages generate more entrepreneurs. A one standard deviation increase in relative wages increases the number of entrepreneurs generated by the average firm by .23, or of approximately a 10 percentage point increase above the baseline. Looking across all columns, the top row in Table 11 shows that firms with better pay generate more entrepreneurs selecting into upstream or anchor industry startups. The triple interactions in the remaining rows show whether the elasticity between wages and employee departures to entrepreneurship varies by supply chain industry. The excluded industry group in the specification is "all other industries." Isolating an individual row and comparing coefficients across columns shows whether firms in that industry grouping yield entrepreneurs disproportionately in certain industries. The rows of Table 11 show higher paying firms generate more entrepreneurs and particularly into their own supply chain category. Upstream (downstream) firms have significantly more entrepreneurs who form new ventures in upstream (downstream) industries than in anchor firm industries or downstream (upstream) industries. Consistent with the suggestive evidence in Table 9 and Table 10 that the number of employee departures from anchor industry incumbent is concentrated in a limited set of firms, Table 11 shows that the firms generating entrepreneurs tend to be higher paying firms.

Using the set of all eventual entrepreneurs in a county, I show that the probability of founding a supply chain firm is increasing in individual ability and industry experience. Table 12 estimates Equation 4 as a linear probability model. The main

Table 11: Number of Employee Departures by Firm Pay

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Win * Mean Wage	0.2324***	0.0100***	0.0016**	0.0014
	(0.0208)	(0.0010)	(0.0007)	(0.0009)
Win * Upstream * Mean Wage	-0.3181***	0.0975***	0.0448***	0.0267*
	(0.0813)	(0.0306)	(0.0134)	(0.0137)
Win * Anchor Industry * Mean Wage	-0.1029	0.2729***	0.2096***	0.1768**
	(0.2528)	(0.0800)	(0.0786)	(0.0740)
Win * Downstream * Mean Wage	-0.0503	0.0119	0.0409***	0.1651***
	(0.0555)	(0.0137)	(0.0114)	(0.0351)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.6369	0.1186	0.0648	0.0851
N	1094000	1094000	1094000	1094000

Notes: * 0.1 ** 0.05 *** 0.01. High paying firms generate the most entrepreneurs across all industries (row 1), though the impact of wage is heterogeneous across industries (rows 2-4). The strongest effects are on spinoffs (e.g., Column 2 Row 2, Column 3 Row 3, and Column 4 Row 4). This table shows the impact firm wages have on employee departures by industry. The coefficients are interpreted as the relative elasticity of employee departures to firm wages by industry. Wages are normalized to have mean 0 and unit standard deviation. Firm wages are defined as average log quarterly earnings for all employees employed in the SEIN-county. Interactions are conducted after standardization to restore main effects. Regressions are weighted by total SEIN-county employment. Standard errors are clustered at the SEIN-county.

interaction in the first row shows that across all supply chain groupings, the probability a founder starts a firm in the supply chain in treated counties increases relative to founding a firm in all other industries. The triple interactions then show the heterogeneous effect of income and being in a winning county by supply chain industry. Row 2 in Table 11 Column 1 shows that a 1 standard deviation increase in wages of an entrepreneur from an upstream industry increases the relative likelihood of starting a firm in an upstream industry by 3%. The difference between this coefficient and the other coefficients related to the row is also statistically different, suggesting higher performing upstream entrepreneurs disproportionately select into upstream entrepreneurship. This pattern generalizes as well for entrepreneurs from anchor industries. However, wages do not appear to have a significant impact on the founding industry of entrepreneurs from downstream employers.

Firms with greater variance in compensation relative to their industry are thought

Table 12: Likelihood Entrepreneur Selects into Industry by Wage

	(1)	(2)	(3)
	Pr(Found upstream firm)	Pr(Found anchor industry firm)	Pr(Found downstream firm)
Win * Wage	0.0047***	0.0017***	0.0022***
	(0.0003)	(0.0002)	(0.0003)
Win * Upstream * Wage	0.0305***	0.0115***	0.0071*
	(0.0087)	(0.0036)	(0.0037)
Win * Anchor Industry * Wage	0.0342***	0.0619***	0.0443***
	(0.0095)	(0.0108)	(0.0126)
Win * Downstream * Wage	-0.0097**	0.0043	0.0111
	(0.0043)	(0.0041)	(0.0097)
Main Effects	Y	Y	Y
Worker Controls	Y	Y	Y
Industry x Year FE	Y	Y	Y
Case FE	Y	Y	Y
Adj. R-Sq.	0.1075	0.0626	0.1519
N	2124000	2124000	2124000

Notes: * 0.1 ** 0.05 *** 0.01. Entrepreneurs in winning counties tend to be higher wage employees relative to their firm of prior employment. This table shows the impact firm wages have on employee departures by industry. The coefficients are interpreted as the relative elasticity of supply chain entrepreneurship to individual wage by industry. Wages are normalized to have mean 0 and unit standard deviation. Firm wages are defined as average log quarterly earnings the eventual entrepreneur's most recent SEIN-county of employment prior to the anchor firm's arrival. Interactions are conducted after standardization to restore main effects. Controls include age, age-squared, sex, race, education, and SEIN fixed effects. Regressions are unweighted. Standard errors are clustered at the SEIN-county.

to be better at rewarding top performers (e.g., Lazear and Shaw 2007). These top performers are more likely to stay in firms that appropriate rents from their skill, but conditional on leaving are also more likely to select into entrepreneurship. Table 13 corroborates this hypothesis and demonstrates that incumbent firms with greater pay dispersion generate more entrepreneurs even outside the supply chain. The evidence on wages aligns with the industry-specific analysis in Section 5.2.1 to support the human capital based hypothesis where the selection of high ability individuals with relevant industry experience in high quality firms into entrepreneurship drives improved startup performance after a local shock to supplier-buyer markets.

The complication related to pre-existing differences driving the results from Section 5.2.1 is also relevant for this section's estimation of Equation 4. To account for this, I compare the profile of spinoffs and entrepreneurs across eventual supply chain industries 8 through 2 years prior to the anchor firm's arrival to the county. I do not

Table 13: Number of Employee Departures by Firm Pay Dispersion

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Win * Wage Dispersion	0.4239***	0.0121***	0.0026***	0.0010
	(0.0360)	(0.0009)	(0.0005)	(0.0011)
Win * Upstream * Wage Dispersion	-0.3974***	0.1124***	0.0390***	0.0282**
	(0.0760)	(0.0282)	(0.0136)	(0.0137)
Win * Anchor Industry * Wage Dispersion	0.5206**	0.4151***	0.4034***	0.2764***
	(0.2309)	(0.0983)	(0.0961)	(0.0927)
Win * Downstream * Wage Dispersion	-0.1233*	0.0159	0.0298***	0.1947***
	(0.0694)	(0.0133)	(0.0112)	(0.0360)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.6375	0.1237	0.0716	0.0868
N	1094000	1094000	1094000	1094000

Notes: * 0.1 ** 0.05 *** 0.01. Firms with greater pay dispersion lose more employees to spinoffs. This table shows the impact firm pay dispersion has on employee departures by industry. The coefficients are interpreted as the relative elasticity of employee departures to firm wage dispersion by industry. Firm wage dispersion is defined as the variance of log quarterly earnings for all employees employed in the SEIN-county. Variance is normalized to have mean 0 and unit standard deviation. Interactions are conducted after standardization to restore main effects. Regressions are weighted by total SEIN-county employment. Standard errors are clustered at the SEIN-county.

find evidence to suggest my results are driven by pre-existing difference in spinoffs between winning and runners-up counties.

5.2.3 Employer's Age and Size

Alternative mechanisms may relate to why aggregate employment does not increase, but instead reallocates to young firms. One such mechanism is that employees depart bureaucratic firms that are unable to take advantage of new opportunities that arise from the anchor firm's opening. A number of studies such as Klepper (2007), Klepper and Buenstorf (2009), and Klepper and Thompson (2010) describe a process where employees who are unable to undertake projects within bureaucratic firms spin-out and pursue projects on their own. These studies point to the canonical examples of clusters formed through these disagreements such as the Detroit automobile cluster and the Akron tire industry. On the contrary, entrepreneurs may take on projects complementary to their former employer (e.g., Oettl and Agrawal 2008). This hy-

pothesis could also be reflected by a business stealing effect. If spinoff employees are also former employees of old large incumbent firms then this could serve as evidence that bureaucratic firms are unable to respond to local supply chain shocks. To test the bureaucracy hypothesis I using size and age as proxies for bureaucracy. I find only inconclusive results, though age and size are crude measures of a firm's bureaucracy.[37]

Large and bureaucratic firms may be less able to undertake new opportunities that arise. Their employees may then spinoff to pursue these projects. This process of the clustering of entrepreneurship has been used to explain the origins of the Detroit automobile cluster and the Akron tire industry (Klepper 2007/10). The microdata do not allow for directly testing for firm bureaucracy specifically or through anecdotal evidence, but do allow for testing the relationship between firm characteristics that may correlate strongly with an incumbent firm's inability to shift and adjust to sudden opportunities.

Tables 14 and 15 test firm age and firm size. The main interaction shown in Table 14 shows that older firms in winning counties generate more entrepreneurs who start firms upstream and anchor industries, though this effect is not particularly economically significant. The triple interactions show the effect does not vary by industry grouping and in some cases older firms spawn more entrepreneurs. Table 15 tests the potential firm size dimension using the linear probability framework and similarly demonstrates mixed evidence on the role of an entrepreneur's prior employer's size and selection into supply chain entrepreneurship.

The mixed evidence related to incumbent firm age and size does not necessarily negate an employee disagreements with employer explanation for the increase in entrepreneurship through spinoffs. Without any contextual evidence, the generic measures of age and size are crude and may mask underlining heterogeneity even

[37]Testing for business stealing is out of the scope of this paper.

Table 14: Number of Employee Departures by Firm Age

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Win * Firm Age	0.0056**	0.0009***	0.0003***	0.0002*
	(0.0028)	(0.0001)	(0.0001)	(0.0001)
Win * Upstream * Firm Age	-0.0196	-0.0004	-0.0029**	-0.0037***
	(0.0120)	(0.0032)	(0.0013)	(0.0014)
Win * Anchor Industry * Firm Age	-0.0336	-0.0126	-0.0080	-0.0144*
	(0.0397)	(0.0081)	(0.0081)	(0.0079)
Win * Downstream * Firm Age	0.0119	-0.0024	-0.0055**	0.0022
	(0.0126)	(0.0028)	(0.0022)	(0.0056)
Main Effects	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.6368	0.1152	0.0604	0.0839
N	1094000	1094000	1094000	1094000

Notes: * 0.1 ** 0.05 *** 0.01. It is unclear whether older firms that may be more bureaucratic lose more employees to spinoffs. This table shows the impact firm age has on employee departures by industry. Firm age is calculated in the LBD as the age of the oldest establishment when the firmid first appears. The firmid then ages naturally. The LBD age measure is then mapped to the LEHD using the firmid-county-EIN-SEIN concordance in the ECF. Interactions are conducted after standardization to restore main effects. Regressions are unweighted. Standard errors are clustered at the SEIN-county.

Table 15: Likelihood of Departure by Firm Size

	(1)	(2)	(3)
	Pr(Found upstream firm)	Pr(Found anchor industry firm)	Pr(Found downstream firm)
Win * Firm Size	0.0014**	0.0005**	0.0008
	(0.0006)	(0.0002)	(0.0005)
Win * Upstream * Firm Size	-0.0136	0.2039***	0.2384***
	(0.1275)	(0.0309)	(0.0339)
Win * Anchor Industry * Firm Size	0.0434	-0.1837***	0.4800***
	(0.1116)	(0.0522)	(0.1239)
Win * Downstream * Firm Size	0.1063	0.0795**	-0.5916***
	(0.0827)	(0.0339)	(0.1126)
Main Effects	Y	Y	Y
Worker Controls	Y	Y	Y
Industry x Year FE	Y	Y	Y
Case FE	Y	Y	Y
Adj. R-Sq.	0.1078	0.0612	0.1530
N	2124000	2124000	2124000

Notes: * 0.1 ** 0.05 *** 0.01. The evidence on the impact of firm size on the likelihood an individual departs into supply chain entrepreneurship is unclear. This table shows the impact of firm size on employee departures by industry. Firm size is calculated in the LEHD sample. Size reflects the number of employees who worked in the firm during the year with an average wage above minimum wage in all quarters worked. Interactions are conducted after standardization to restore main effects. Notes: * 0.1 ** 0.05 *** 0.01. Entrepreneurs in winning counties tend to be higher wage employees relative to their firm of prior employment. This table shows the impact firm wages have on employee departures by industry. The coefficients are interpreted as the relative elasticity of supply chain entrepreneurship to individual wage by industry. Wages are normalized to have mean 0 and unit standard deviation. Interactions are conducted after standardization to restore main effects. Controls include age, age-squared, sex, race, education, and SEIN fixed effects. Regressions are unweighted. Standard errors are clustered at the SEIN-county.

within those characteristics. The results in Sections 5.2.1, 5.2.2, and 5.2.3, however, remain consistent with a view supporting the general role of proximity to buyer and supplier markets for entrepreneurial formation and performance alongside individuals with location and industry specific human capital being most able to capture gains to local industry shocks. Future research should probe at understanding whether incumbent firms face a business stealing effect from employee departures or if departing entrepreneurs form businesses in the same supply chain category, but complementary or non-competing product markets.

5.3 Within-Firm Wage and Occupation Effects

I examine distributional impacts by taking the set of winning counties and measuring the treatment effect of the anchor firm's arrival within the set of winning counties. Specifically, I compare the characteristics of spinoffs that form between years $t-8$ and $t-2$ with those that form in the defined post anchor firm period. Whereas prior analyses focused on the direct effects of the anchor firm's arrival on employee departures from incumbent firms and the success of those departing, this section turns towards policy implications of anchor firms arriving to a county. The key features are that the introduction of an anchor firm does not increase aggregate employment in supply chain industries, but simply employment in young firms. The young firms that form in a counting become much more likely to be founded by supply chain entrepreneurs from top paying firms.

Table 16 shows that incumbent firms in winning counties lose more employees to entrepreneurship after the anchor firm arrives. Consistent with Section 5.2.1, entrepreneurs become more likely to come from supply chain related industries that the set of all other industries. Within the supply chain, the composition of firms who generate entrepreneurs in the supply chain also shifts towards incumbent supply

chain industries. Upstream incumbents generate .18 more spinning out employees than prior to the opening, an effect that represents about a 30% increase above the baseline spinoff rate or two spinoffs for every 10 business units in the county. Downstream incumbents similarly generate 4 new spinoffs for every 10 extant business units. Table 16 shows that the comparisons between winning and runners-up counties on the dimension of industry background are driven by changes in spinoff composition within winning counties. Though currently suppressed, entrepreneur-level regressions show qualitatively similar and strong effects of founder experience on selection into supply chain entrepreneurship.

Table 16: Change in Departures by Parent Firm Industry

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Post * Upstream	-0.0389	0.1797***	0.0447***	0.0371***
	(0.0858)	(0.0263)	(0.0122)	(0.0124)
Post * Anchor Industry	1.294***	0.4705***	0.5257***	0.2959***
	(0.2601)	(0.1082)	(0.1004)	(0.0998)
Post * Downstream	0.3380***	0.1035***	0.0880***	0.3918***
	(0.1094)	(0.0281)	(0.0235)	(0.0550)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.5483	0.1611	0.0793	0.0932
N	332000	332000	332000	332000

Notes: * 0.1 ** 0.05 *** 0.01. The anchor firm's arrival induces spinoffs across the supply chain. Incumbent supply chain firms lose employees to entrepreneurship in the supply chain, but the strongest effects are on spinoff entrepreneurship. For example, upstream firms lose more employees to upstream entrepreneurship (row 1). This table displays coefficients of interest from estimating Equation 3 restricted to the set of winning counties and replacing the *Win* dummy variable with a *Post* dummy variable. This table tests whether the industries that yield spinoffs after the anchor firm arrives changes in winning counties. The sample includes only those employees whose average quarterly earnings across quarters worked in the firm exceed $2,678 which represents earnings for a minimum wage employee working 40 hours for 13 weeks. Wages are then inflated to 2015 dollar amounts. The regressions include all main effects and interactions shown in Equation 3, but the table displays only key coefficients of interest. Firm controls include controlling for firm size and firm age. Industry by year fixed effects control for secular trends in startup formation that vary by industry. Case fixed effects ensure that comparisons are made between firms within each anchor firm event. The case identifier is interacted with a dummy variables taking the value of 1 if the year is between 5 years prior to the anchor firm event through 7 years after. This facilitates identification of the industry year fixed effects while ensuring the time periods where loading on the difference-in-differences estimator is bounded.

A number of recent studies have examined the causes and effects of increased sorting of high ability workers into top paying firms (Barth et al., 2016; Card et al., 2018; Song et al., 2019). This study considers the role top paying firms have on incubat-

ing employees who become primed to seize opportunities that suddenly arise perhaps through observing shocks that arise in their local market, but outside their direct industry. Though currently undisclosed, loading on the difference-in-differences estimator is largely driven by the earliest entry cohorts–employees inside the county with substantial industry experience prior to the anchor firm's arrival. These individuals are in fact top earners in high quality firms which may further suggest that sorting does not only improve labor outcomes within firm-worker matches, but also through a channel whereby top performing employees are more able to seize an outside option of entrepreneurship rather than wage-earning employment.

Table 17: Change in Departures by Pay of Parent Firms

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Post * Mean Wage	0.2645***	0.0092***	0.0014*	0.0021**
	(0.0130)	(0.0012)	(0.0008)	(0.0008)
Post * Upstream * Mean Wage	-0.0801	0.1127***	0.0785***	0.0678***
	(0.0605)	(0.0288)	(0.0171)	(0.0173)
Post * Anchor Industry * Mean Wage	-0.0432	0.2072**	0.1866**	0.1842**
	(0.2265)	(0.0826)	(0.0785)	(0.0731)
Post * Downstream * Mean Wage	-0.0634	0.0304	0.0509***	0.1699***
	(0.0797)	(0.0198)	(0.0154)	(0.0378)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.5492	0.1640	0.0867	0.0988
N	332000	332000	332000	332000

Notes: * 0.1 ** 0.05 *** 0.01. Firms that pay employees more than their industry competitors generate more entrepreneurs after the anchor firm's arrival to the county. The magnitude of the effects varies by industry with anchor industry incumbents and downstream incumbents losing the most. This table shows coefficients of interest from estimating Equation 4 restricted to the set of winning counties and replacing the Win dummy variable with a $Post$ dummy variable. This specification tests whether the elasticity of spinoffs to firm mean wages varies by industry. Mean wages is measured as the average worker's average quarterly earnings in a SEIN. The sample includes only those employees whose average quarterly earnings across quarters worked in the firm exceed $2,678 which represents earnings for a minimum wage employee working 40 hours for 13 weeks. Wages are then inflated to 2015 dollar amounts. The mean firm wage is transformed to have mean 0 and unit standard deviation. The regressions include all main effects and interactions shown in Equation 4, but the table displays only key coefficients of interest. Firm controls include controlling for firm size and firm age. Industry by year fixed effects control for secular trends in startup formation that vary by industry and imply that firm wages are relative to other firms in their own industry. Case fixed effects ensure that comparisons are made between firms within each anchor firm event. The case identifier is interacted with a dummy variables taking the value of 1 if the year is between 5 years prior to the anchor firm event through 7 years after. This facilitates identification of the industry year fixed effects while ensuring the time periods where loading on the difference-in-differences estimator is bounded.

I show that the composition of supply chain entrepreneurs changes when an anchor firm arrives by the growth rate of wages in the firms economically proximate industries

to the anchor firm. These employees are typically above the mean wage-earners in their firm, are not considered to be the top wage earners in their firms. Table 17 shows that high wage growth firms increase the number of employee departures into entrepreneurship. Table 18 shows that supply chain industries also strongly correlate with occupational similarity. The table shows that the correlation across all industries represents a shift in the earnings profile of entrepreneurs in the county. Table 18 follows Table 13 in demonstrating that firms in the supply chain that are considered to be better at compensating top performers within the firm do not necessarily yield more entrepreneurs overall but do generate more entrepreneurs who select into the supply chain. Entrepreneur-level results show that the main effect on the $Post \times Wage$ variable is positive and strongly significant on regressions estimating the likelihood of selecting into a supply chain industry. The main effect does not vary by industry of a founder's origin.

This paper posits that this result corresponds to layer shedding in the firm (Caliendo, Monte, and Rossi-Hansberg 2015).[38] Strikingly, the results continue to show that the effects are specific to supply chain firms who disproportionately contribute to the number of employees spinning out to found other supply chain firms. Tables 17 and 18 fix the log wage pre-anchor arrival and display log wages after the anchor firm arrives. This means that the coefficients represent a log wage growth relative to pre-arrival. The first row of Table 17 shows that the combination of wage growth firms generating more entrepreneurs as well as overall industry employment remaining constant suggests a reallocation of workers across firms. If workers are displaced from their jobs, it may be the case that they simply transition into former occupations in industries that disproportionately hire those same occupations.[39] Lazear and Oyer

[38] Greenstone, Hornbeck, and Moretti (2010) show that wages increase in industries treated by the anchor firm.

[39] Firms facing the greatest exogenous shock to wages are also in the most high-tech industries

Table 18: Change in Departures by Pay Dispersion of Parent Firms

	(1)	(2)	(3)	(4)
	# of employees founding a firm	# of employees founding upstream firm	# of employees founding anchor industry firm	# of employees founding downstream firm
Post * Wage Dispersion	0.5397***	0.0131***	0.0019***	0.0029***
	(0.0178)	(0.0011)	(0.0007)	(0.0008)
Post * Upstream * Wage Dispersion	-0.0834	0.1446***	0.0722***	0.0666***
	(0.0622)	(0.0274)	(0.0170)	(0.0171)
Post * Anchor Industry * Wage Dispersion	0.6352***	0.3992***	0.3881***	0.3445***
	(0.2327)	(0.1247)	(0.1169)	(0.1132)
Post * Downstream * Wage Dispersion	-0.1179	0.0384*	0.0588***	0.2348***
	(0.0738)	(0.0208)	(0.0161)	(0.0370)
Main Effects	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y
Industry x Year FE	Y	Y	Y	Y
Case FE	Y	Y	Y	Y
Adj. R-Sq.	0.5530	0.1695	0.0966	0.1063
N	332000	332000	332000	332000

Notes: * 0.1 ** 0.05 *** 0.01. The top row shows that incumbent firms with greater pay dispersion lose more employees to entrepreneurship across a broad set of industries. The main effect shows that a one standard deviation increase in wage dispersion leads to approximately a 30 percentage point increase in the number of employee departures. This magnitude is similar for changes in the number of employees who spinoff from upstream, anchor industry, and downstream firms. This table shows coefficients of interest from estimating Equation 4 restricted to the set of winning counties and replacing the Win dummy variable with a $Post$ dummy variable. Wage dispersion is measured as the variance in SEIN's average worker quarterly wages. The sample includes only those employees whose average quarterly earnings across quarters worked in the firm exceed $2,678 which represents earnings for a minimum wage employee working 40 hours for 13 weeks. Wages in this cleansed sample are inflated to 2015 dollar amounts. The variance in wages is transformed to have mean 0 and unit standard deviation. The regressions include all main effects and interactions shown in Equation 4, but the table displays only key coefficients of interest. Firm controls include controlling for firm size and firm age. Industry by year fixed effects control for secular trends in startup formation that vary by industry. Case fixed effects ensure that comparisons are made between firms within each anchor firm event. The case identifier is interacted with a dummy variables taking the value of 1 if the year is between 5 years prior to the anchor firm event through 7 years after. This facilitates identification of the industry year fixed effects while ensuring the time periods where loading on the difference-in-differences estimator is bounded.

(2004) and Lazear and Shaw (2007) discuss that wage pressures on firms is largely driven by external local industry factors. Combined with Caliendo et al., given that compensation is a key component of employee retention across occupational sectors, growing firms that shed mid-management layers may also be those that generate the most employees who select into firms that complement their own skills.

I use the Bureau of Labor Statistics (BLS) Occupational Employment Statistics (OES) to provide descriptive evidence that supply chain industries are also occupationally similar industries. The OES creates occupational codes that are constant across industries and provides the number of employees that are accounted for in each occupation. I tabulate the percentage of industry employees in each occupation and compare the occupational share of industry across anchor and supply chain industry. Table 19 uses the industry pairs in Table 3 and 5 to show the pairwise correlations between occupational shares of those industries.

Table 19: Occupational Similarity for Top Supply Chain Pairs

Announcement industry	I/O industry	Occupational correlation
3222. Converted Paper Product Manufacturing	3221. Pulp, Paper, and Paperboard Mills	0.9396
3361. Motor Vehicle Manufacturing	3362. Motor Vehicle Body and Trailer Manufacturing	0.9228
3361. Motor Vehicle Manufacturing	3363. Motor Vehicle Parts Manufacturing	0.9054
5250. Funds, Trusts, and Other Financial Vehicles	5239. Other Financial Investment Activities	0.8008
3252. Resin, Synthetic Rubber, and Artificial Synthetic Fibers and Filaments Manufacturing	3251. Basic Chemical Manufacturing	0.6363
4830. Water Transportation	3366. Ship and Boat Building	0.5119
3252. Resin, Synthetic Rubber, and Artificial Synthetic Fibers and Filaments Manufacturing	3261. Plastics Product Manufacturing	0.4158
3241. Petroleum and Coal Products Manufacturing	2110. Oil and gas extraction	0.3469
3116. Animal Slaughtering and Processing	1123. Poultry and egg production	0.1993

Notes: The table shows similarity of occupational composition in the leading anchor industry supply chain pairs in terms of percentage of all input/output to/from the paired industry. Occupational composition is tabulated using the Bureau of Labor Statistics' Occupational Employment Statistics (OES). The OES provide occupational codes and the share of industry employment in each code. This table displays between industry correlations of the percentage of industry workers employed in an occupation. The table shows that the share of an industry's shipments to/from an industry correlate strongly with the occupational mix in those industries. Overall, the correlation of occupational shares between anchor firm industries and their supply chain industries is 0.2894. By comparison, the correlation of occupational mix in anchor industries and the full set of industries is 0.0598. Predictably industries within the same sector have similar occupational mixes as well. Within the same 3-digit NAICS designations, the correlation is 0.4891. Within 3-digit NAICS and within the supply chain the correlation is 0.6673. The table excludes shipments from an industry to itself.

The table shows that supply chain linked industries display stronger occupational similarity than all other industry pairs. The most intensely related anchor firms and paired industries have a pairwise occupational share correlation of 0.2894. This

where the supply of labor may be relatively inelastic by occupation.

correlation is significantly different from the correlation between the full set of industries (0.0598). As displayed, many of the paired supply chain industries are within the same 3-digit NAICS sector which may mechanically yield stronger correlations by virtue of being already classified as similar industries. The correlation between industries of the same NAICS-3 sector is 0.4891 which either implies a mechanical relationship or shows that even within the set of neighboring industries, supply chain industries also share a common occupational component. Taking the set of 3-digit NAICS industries, the occupational composition between the anchor firm and supply chain industries is a significantly stronger correlation of 0.6673. These correlations suggest that the proper interpretation of the upstream and downstream designations combined with the wage effects suggest that labor is shared between *economically proximate* industries in the anchor industries.

Table 20 shows the most correlated industry supply chain pairs in the anchor firm sample. The table combined with Table 1 shows that the most represented industries on which most loading occurs in the difference-in-differences estimator is by anchor industry pairings most occupationally similar.

Table 20: Leading Supply Chain Pairs by Correlation

Announcement industry	I/O industry	Occupational correlation
3342. Communications Equipment Manufacturing	3345. Navigational, Measuring, Electromedical, and Control Instruments Manufacturing	0.9716
3369. Other Transportation Equipment Manufacturing	3362. Motor Vehicle Body and Trailer Manufacturing	0.9639
3119. Roasted Nuts and Peanut Butter Manufacturing	3114. Fruit and Vegetable Preserving and Specialty Food Manufacturing	0.9467
3115. Dairy Product Manufacturing	3114. Fruit and Vegetable Preserving and Specialty Food Manufacturing	0.9400
3221. Pulp Mills	3222. Converted Paper Product Manufacturing	0.9400
3222. Converted Paper Product Manufacturing	3221. Pulp Mills	0.9400
3369. Other Transportation Equipment Manufacturing	3363. Motor Vehicle Parts Manufacturing	0.9348
3115. Dairy Product Manufacturing	3119. Roasted Nuts and Peanut Butter Manufacturing	0.9284
3115. Dairy Product Manufacturing	3112. Grain and Oilseed Milling	0.9283
3361. Leather and Hide Tanning and Finishing	3362. Motor Vehicle Body and Trailer Manufacturing	0.9228

Notes: See Table 19. The table displays the Top 10 correlations between the anchor industry and any one of its supply chain pairs.

These findings together reveal that the arrival of an anchor firm does not result in an aggregate rise in entrepreneurship in a county through a discovery of new opportunities. Firm entry is specific to economically proximate, as measured by input-

output intensity and occupational similarity, and by workers who seem to transition between those industries when pushed out of incumbent firms. Entry within the supply chain is not balanced across individual characteristics such as worker skills or experience, but may be driven by differences in firm-specific effects as evidenced by wages and personnel decisions firms make after the anchor firm arrives. Additional work, however, may provide further evidence on the differences between effects on entrepreneurial opportunity and a layer shedding effect. This paper provides evidence that extant firms face rising factor prices through labor market channels. However, this paper does not test interactions between firm size and firm age, nor do I explicitly examine the extent to which departing founders employ lower wage employees from their prior employer.

6 Robustness

6.1 Placebo Tests

The main checks testing the validity of the empirical strategy related to Section 5.1 include directly testing for parallel trends using a dynamic difference-in-differences estimation and conducting placebo tests. The parallel trends test uses time interactions for each year between five years prior to the event and seven years post event. In these regressions, there is no evidence to reject the null hypothesis that the coefficients $win \times time$ interactions for periods prior to the announcements are 0. The dynamic panel also allows for computing the specific time periods in the post-event interval where most of the loading occurs on the difference-in-differences estimators. I find that the first three years of entry cohorts and founders drive the impacts in new venture creation and performance. This also tends to be the period during which top earners in local incumbent firms spinout from their employers.

Placebo tests ensure that the coefficients of interest on the difference-in-differences estimators and interactions are driven by the treatment of the specific case's winning county and its counterfactual county. The first placebo test randomizes the treated county within each set of case counties.[40] A second set of placebo test randomizes the treatment industry within each case's contest counties.[41] The lack of significance on coefficients of interest support that the differences are driven by the anchor firm's arrival in the winning county.

The analysis on spinoffs from incumbent firms and entrepreneurs compares subsequent outcomes of spinoffs and entrepreneurs in the winning and runners-up counties in year $t-2$ over the first 7 years of the anchor firm's arrival to the county. The spinoff process may be inherently different between winning and runners-up counties and the interactions of the variables of interest with the Win dummy variable may reflect these innate differences. Using work histories and spinoffs from years prior to the anchor firm's arrival to a county, I do not find evidence there are inherent differences between the counties *ex ante* in terms of selection into spinoffs or entrepreneurship.

6.2 Subsidies

My identification strategy assumes that profit maximizing firms select locations that minimize operating costs associated with the its production function. Therefore, the anchor firm's shortlisted sites is assumed to reflect locations that deliver equivalent operating costs with respect to its production function. However, subsidies may allow a firm to consider a location with lower profitability if the opportunity cost is compensated by the locale using tax incentives and other support mechanisms.

[40] Suppose Case 1 involves counties A and B. The placebo tests randomly selects either A or B as the winning county. This randomization is repeated over all cases.

[41] For each case's counties, the placebo test randomizes the industry the "anchor firm" opens, thereby creating a fictitious opening event.

The prevalence of subsidies is non-trivial. During 2015, state and local governments spent an estimated $45 billion in incentive programs targeting specific companies (Bartik 2017). Using publicly available GoodJobsFirst data and information in *Site Selection* magazine, I collect data on subsidy programs for site searches in the full *Site Selection* sample.[42] Obtaining information on competing bids that lost the contests is difficult because these jurisdictions do not actually disburse funds and further avoid disclosure to media and government spending watchdog organizations. In cases where magazine articles or the company's own public statements describe the bidding process for counterfactual counties, I record the subsidy types and amounts. I find that subsidies are almost universal in the site searches that *Site Selection* magazine covers. Perhaps interestingly, winning and runners-up counties offer similar packages with the firm's ultimate location choice driven by idiosyncratic preferences. In one case, the magazine points to a costly site search that ended when a C.E.O. "in an unusually strongly worded comparison of the two contenders... mentioned average temperatures" as the deciding factor. Site searches the magazine describes under section headings "Top Deals" or "Megadeals" often pertain to multibillion dollar events that are more common for large auto and aircraft production facilities. The size of the incentive programs increase in the capital intensity of the project. Aircraft and auto manufacturing facilities command the largest subsidies, though counterfactual counties offer competitive packages. In research and development facilities, incentives appear to play a less important role relative to the presence of universities, existing corporate labs, and a skill-specific labor pool. Companies opening these types of facilities even refuse incentive packages and it is not uncommon for the magazine to opine that the C.E.O. of a firm opening a knowledge-intensive facility turned down favorable offers for cities that offered no incentives at all.

[42]The firms in the subsidy sample is not based on matches to administrative Census data.

Other location-specific factors unobserved to the econometrician that simultaneously affect the entrepreneurial environment may also drive the anchor firms' site search. Firms may consider sites where they project to have greater influence over future laws and regulations they face. Political economy considerations aside, firms may also anticipate follow-on activity by large established suppliers or customers who will collocate. If this is true, then my results on entrepreneurship may be responses to follow-on anchor firms and not the initial firm I attribute my results to. For example, news articles surrounding a site search for a large manufacturing plant covered by the magazine discuss the formation of a follow-on cluster populated new facilities of the firm's existing input suppliers and output My results for entrepreneurship are driven by the earliest years after the anchor firm locates to a county whereas increases in the number of large firms and establishments does not occur until the last years of a case's time horizon.

6.3 Identifying Spinoffs in the LEHD

In the final LEHD sample, I identify an entrepreneur as a top 3 wage earner in a new *SEIN* associated with a *firmid* aged less than three years of age.[43] Firm characteristics on SEINs are obtained by merging establishment-level data in the Longitudinal Business Database summed to the federal EIN-county level since this is the narrowest observational unit common to the LBD and LEHD. Matching on EIN-county and importing age based on the *firmid*'s age, I obtain a list of matched SEIN-counties. I take SEIN-counties associated with SEINs that have never appeared before in the data as new firms.

[43]Wages are defined by the average quarterly wage associated with a Personal Identification Key (PIK) in non-zero wage quarters associated with a SEIN-SEINUNIT. The sample excludes jobs for which the average quarterly wage is less than full-time minimum wage ($5.15 * 40 hours/week * 13 weeks). After cleansing, wages are adjusted to 2015 dollar amounts.

Spinoffs are formed when a new SEIN operates in the same 4-digit NAICS code as the founder's most recent employer. Parent firm and spinoff size are tabulated using the Employer Characteristics File in the LEHD with robustness checks using the number of individuals in the cleansed LEHD sample. New firms are defined as firms aged less than three years as in the Longitudinal Business Database cleansing procedure. Because a firm in year $t-1$ could appear as a "startup" in year $t+2$ in terms of age, I require a founder to operate in a different SEIN post-announcement than in the pre-period.

6.4 Alternative Measures of Supply Chains

The current measure of supply chains does not include all value enhancing activities associated with a focal industry. In robustness checks, I use NAICS-4 industries with the same NAICS-3 codes, but different NAICS-4 designations to signify "adjacent industries." Often, these industries are complementary to the focal NAICS-4 industry. My results are qualitatively similar, if not stronger, to including adjacent industries as the set of supply chain industries. This alternative measure similarly does not capture all value enhancing industries. For example, video games (software) would not be included as part of the supply chain of a video game console (manufacture) firm. However, constructing a complete set of complementary activities will require a more contextualized and refined industry analysis for future research.

An anchor firm's industry may disproportionately source inputs or ship outputs to a startup's industry, but the startup's dependence may be relatively small. To account for this, I include interaction terms in Equation 1 for the share of a startup firm's industry inputs (output) from (to) the anchor firm industry. I do not find evidence of heterogeneity within the set of upstream and downstream firms by dependence of the startup industry. In other robustness checks, I include all industries without

defining a subset of upstream and downstream industries and a continuous measure of input/output shares to and from industries as an interaction with the treatment variable. I find results qualitatively similar to the main findings on the importance of input-output linkages in startup formation and spinoffs.

Glaeser and Kerr (2009) provide alternative continuous measure that describe the availability of inputs and the local market size for output. The input measure sums the difference between the share of an industry i input from industry k ($Input_{i \leftarrow k}$) and the share of the county's total employment in industry k.

$$Input_{ict} = - \sum_{k=1,...,K} \Big| Input_{i \leftarrow k} - \frac{E_{kct}}{E_{ct}} \Big|, \tag{7}$$

Larger values (closer to 0) correspond to the county's employment across industries being in proportion to the share of total inputs used by a focal industry. The output measure is constructed to capture the ability of an industry to sell locally.

$$Output_{ict} = \Big[\sum_{k=1,...,K} Output_{i \rightarrow k} \frac{E_{kct}}{E_{kt}} \Big] \Big[\sum_{k=1,...,K} Output_{. \rightarrow k} \frac{E_{kct}}{E_{kt}} \Big]^{-1} \tag{8}$$

As before, $Output_{i \rightarrow k}$ is the share of i's output to k. This term is multiplied by the share of industry k's employment in county c ($\frac{E_{kct}}{E_{kt}}$). The first bracketed term effectively interacts output intensity to an industry the degree of localization of that industry.[44] This interaction is scaled by the second bracketed term that measures the total size of the output market. I construct these measures for the year prior to the anchor firm's announcement and tabulate changes in the measure post-anchor firm arrival. These county-industry level measures are similar between winners and runners-up counties prior the anchor firm's announcement.

[44] The degree of localization is somewhat reflective of how tradeable the sector's goods are (Ellison and Glaeser 1997).

6.5 Loading on Time Periods and Events

A growing body of literature critiques event-study difference-in-differences designs. Borusyak and Jaravel (2017) and Goodman-Bacon (2018) discuss estimation issues when subjects are treated at different times. The variation in timing may lead to irregular and inconsistent weighting of events in the difference-in-differences estimator. I conduct preliminary checks on disproportionate loading by time periods and events. I find that loading on the difference-in-differences estimator is driven timing within each event and the industry of the event rather than the years of the individual events or counties themselves.

I test entry cohort effects to capture the persistence of the anchor firm's arrival shock over time. I construct entry cohorts as firms aged 0 in year $t-k$ or $t+k \; \forall \; k$ relative to the opening announcement. I compare startups in winners and runners-up county in terms of cumulative employment growth for each year through 5 years after the startup cohort enters. I estimate the following modified empirical response function:

$$\log\left(\frac{emp_{fjct}}{emp_{fjc0}}\right) = \alpha_0 + \theta_t \times Win_{jc} + \gamma_j + \gamma_c + \varepsilon_{fijct} \tag{9}$$

for each $\theta_t \in (t, t+5]$. The strongest effects on employment growth in winning counties are found over the first three years of the startup's formations. This structure also allows me to compare employment growth rates for startups formed in each year since $t-5$. I refer to each of startups formed in a year t as an entry cohort. The findings suggest the entry cohorts in the first four years after the anchor firm's arrival announcement drive the startup activity documented in the main findings of the paper. I interpret this result to lend credibility to the importance of localized knowledge by specialists already in the county. After several years elapse, general

equilibrium effects may dominate and moderate subsequent entrepreneurial activity. I leave these longer-term adjustments in the county to future research.

In other checks, I group events by timing (e.g., announcements between 1990 and 1994, 1995-1999, 2000-2004, etc.) as well as industry groupings (e.g., Bureau of Labor Statistics defined high-technology categories). The strength of startup formation, entry size, and employment growth effects are qualitatively similar across timing groups, but not industry groupings. Events associated with the high-tech sector appear to have the strongest effects in a manner consistent with literature on local employment multipliers Moretti (2010).

7 Conclusion

I use the geographic expansion of a large firm through openings of large establishments in new counties as a shock to geographic proximity of potential buyers and suppliers to local firms and potential entrepreneurs. This paper shows that the process of spinoffs can be driven by shocks outside an incumbent firm's industry. In particular, a shock to the firm's local supply chain either motivates employees to depart their employer or forces extant firms to streamline and shed layers in the face of rising labor costs. Firms in the anchor firm's industry as well as industries that purchase from or supply to the anchor's industry face the sharpest rise in wages. Incumbent firms already in a county-industry pair of the anchor's supply chain lose employees to entrepreneurship either because these employees perceive a new opportunity due to the anchor firm's proximity or because incumbent firms reorganize. These new entrepreneurs disproportionately found new ventures in the same industry as their former employer, suggesting a role of local industry experience. Incumbent firms in industries outside the anchor firm's supply chain and who do not face rising labor

costs do not experience an increase in employee departures to entrepreneurship. The overall startup entry rate does not increase in the county, but instead the anchor firm's arrival alters the industry composition of entrepreneurship.

My findings are consistent with theories of startup performance in clusters driven by both proximity to buyers and suppliers as well as specialized human capital. These forces dictate which employees are primed to seize opportunities that arise by a sizable shock through the arrival of a firm upstream or downstream to their employer. Founders with supply chain experience disproportionately enter entrepreneurship in the supply chain and these effects are magnified for higher wage employees from high paying firms. These large local industry shocks, however, do not induce a mass of entry by a broad set of individuals with diverse experience or through migration. Instead, individuals from firms most affected by rises in local labor costs are most likely to depart the firm and found their own venture. Whether these individuals are explicitly capturing an opportunity incumbents are unable to capture or are being pushed out of incumbent firms and into smaller, young firms of their own is less clear. However, the rise in supply chain wages and stability in employment support predictions from a layer shedding view of incumbent firms. Because these departing workers have local industry knowledge, they are more likely to work in firms that best utilize that knowledge. Therefore, it is not surprising that my effects are strongest in industries that require the most specialized knowledge and where labor supply may be relatively more inelastic.

This paper connects the literature describing the clustering of input-output industries to literature on management practices and productivity spillovers from anchor firms. (Ellison, Glaeser, and Kerr, 2010) show that coagglomeration patterns are particularly strong for industries that purchase from and supply to each other. While I show a clustering of new firms about an anchor firm are linked through input-output

industries, startup formation most likely occurs because rising wages and labor costs in industries most directly affected by the anchor firm's arrival are forced to streamline operations and potentially shed extra layers of management. These workers then use their industry-specific skill to start firms in the same industry as their employers or industries that complement their accumulated skills. This process is particularly strong in more knowledge intensive industries such as RD services and high-tech manufacturing where local labor supply is relatively more inelastic relative to less knowledge intensive industries such as textile manufacturing.

The determinants of wages in the short-run are determined mostly by industry and location factors external to the firm (Lazear and Oyer 2004). In responding to a sudden and unexpected rise in labor costs, firms may respond to their external environment by reorganizing. While the data does not allow me to determine whether firms are explicitly reorganizing by adding and dropping layers (Caliendo, Monte, and Rossi-Hansberg 2015) or employees willingly depart, firms exhibiting fastest wage growth after the anchor firm's arrival also shed the most workers who form spinoffs. These workers come from higher paying firms, but are not disproportionately the top earners within their firms. These employee transitions in response to shocks to the prevailing wage may help explain findings in prior literature exploiting a similar empirical framework. For example, Bloom et al. (2019) show that management practice scores increase for extant firms when the anchor firm arrives. They attribute their finding to managers learning from the arriving firm. This paper suggests a specific channel through which managers improve: The anchor firm forces incumbents to reorganize and shed less productive layers. These changes may be reflected in structured management practice scores. Similarly, better management practices through reorganization and layer shedding also would be reflected in measured labor productivity improvement in surviving incumbent firms found by Greenstone, Hornbeck, and

Moretti (2010).

This paper's finding that the anchor firm's arrival induces entry and employment in segments of the supply chain, but not across wide industry groups is consistent with prior literature assessing the impacts of subsidies and place-based policies. For example, Moretti (2010) summarizes research on local economic multipliers and concludes that dynamic effects may offset employment gains in one industry by reducing employment in another and that job gain multipliers are largest in knowledge intensive and high-technology sectors. In the ballpark of Moretti's (2010) estimate of targeted place-based policies, a back of the envelope calculation places the local multiplier at 1.3 jobs created for every anchor firm job. While I discuss agglomeration channels, similar to Glaeser and Gottlieb (2008) and Glaeser and Gottlieb (2009) I cannot prescribe a policy solution or optimal subsidy strategy without establishing a parametric form to agglomeration spillovers. However, the aggregate effects appear relatively small given subsidy amounts that often exceed $100 million.

This study must be qualified as it abstracts away from two key components related to the shock in the difference-in-differences and direct testing of the knowledge mechanism. The empirical strategy and interpretation assumes the only shock occurring in the county is the arrival of the anchor firm. However, anchor firms may locate in a county with the expectation of the arrival of other large suppliers. These suppliers may be large established firms who also expand into the county for the first time following the anchor firm. The startup response I observe may then be linked to these follow-on entrants. I also do not match the location of universities and R&D labs to counties where anchor firms arrive. Future research will explore both dimensions using the Longitudinal Business Database to identify follow-on expansions into the county and the Business R&D and Innovation Survey (BRDIS) for information of the location of research facilities. More work must also be conducted that considers the

endogenous matching of workers and firms to locations with the highest productivity. Unobserved selection of firms that incubate entrepreneurs or matching of anchor firms to winning locations with an expected entrepreneurial climate or improving economic conditions may also explain at least some of the results. While the balance tests and various checks on difference-in-differences assumptions may assuage these concerns, the interpretation of the results and agglomeration channels may point to future discussions on labor pooling and variation in worker-firm matches across locations.

A research stream emanating from this paper relate to understanding how labor market effects vary with the industry of the anchor firm. The effects on incumbent wages should vary by how specialized labor is in those industries. Incorporating measures of knowledge intensity of the industry, innovation intensity of the firm, and exploiting information on the location of corporate labs in the BRDIS may provide additional evidence to highlight the labor-specific channels suggested in this paper. Though the evidence presented in this paper is consistent with views on firm reorganization, additional work should be done to specifically identify whether firms are eliminating layers from their organizations. This paper also stops short of discussing welfare implications. The jobs per subsidy dollar seems unreasonably high given the localized nature of the treatment effect, but it is unclear whether incumbent firm reorganization and the movement of labor towards younger firms provides other welfare gains. Understanding the longer-term dynamics of these startups and production in these counties may provide a more complete picture on the welfare implications of these anchor firm arrivals.

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